

BFS Traversal

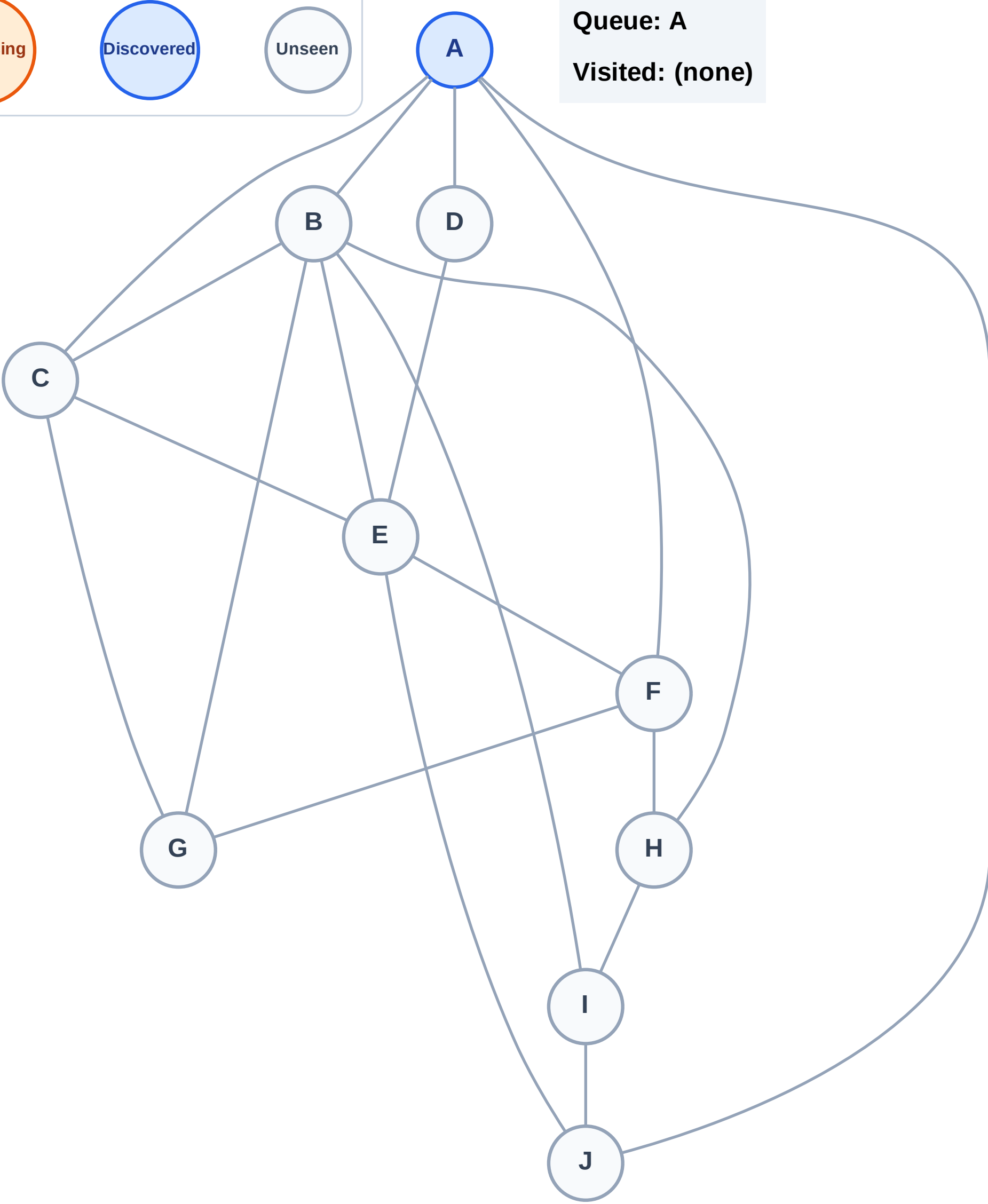
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

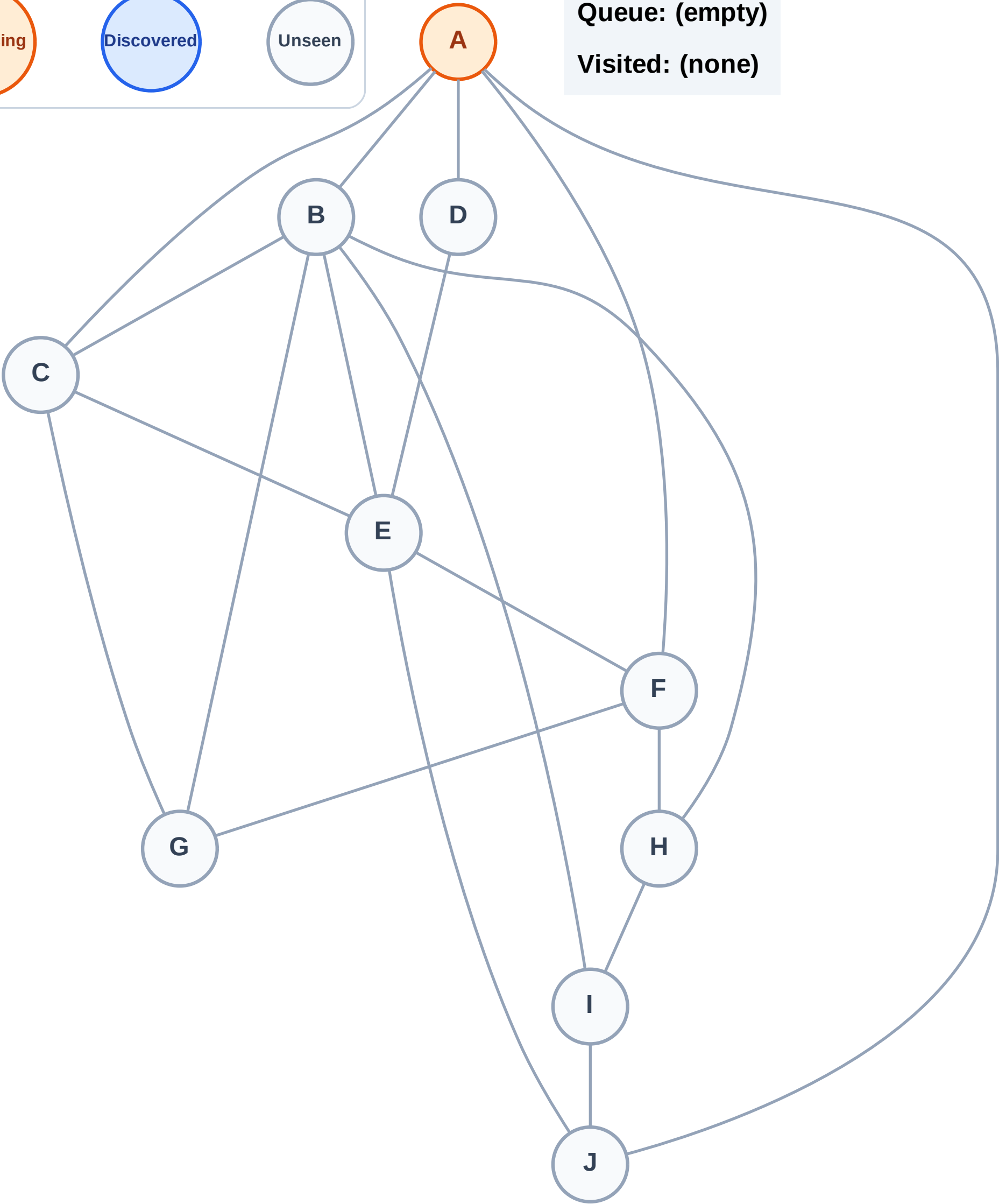
Complete

Processing

Discovered

Unseen

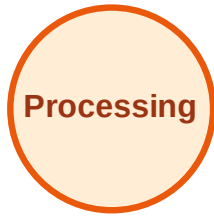
Queue: (empty)
Visited: (none)



BFS Traversal

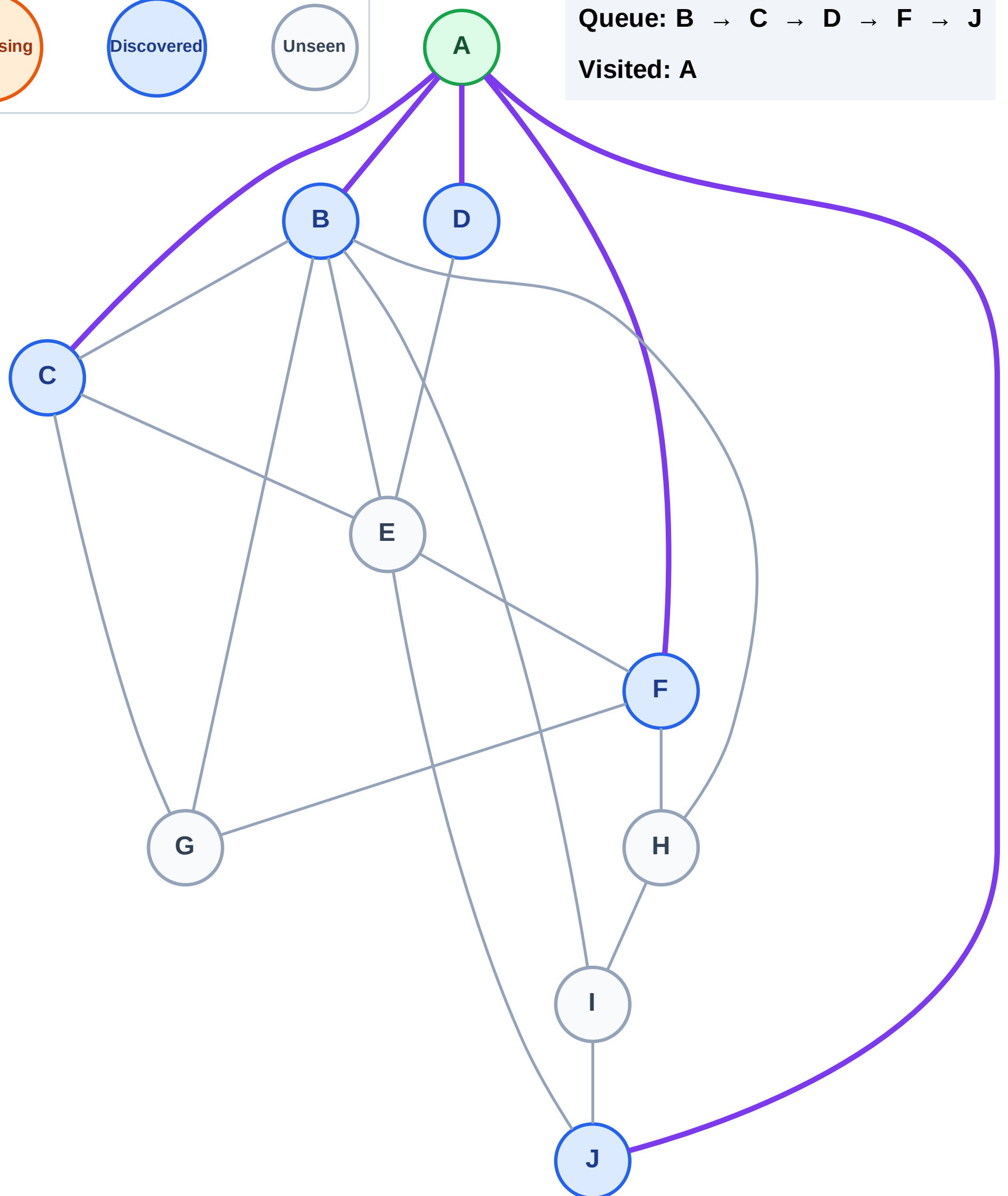
Step 3: Discover B, C, D, F, J from A

Legend



Queue: B → C → D → F → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

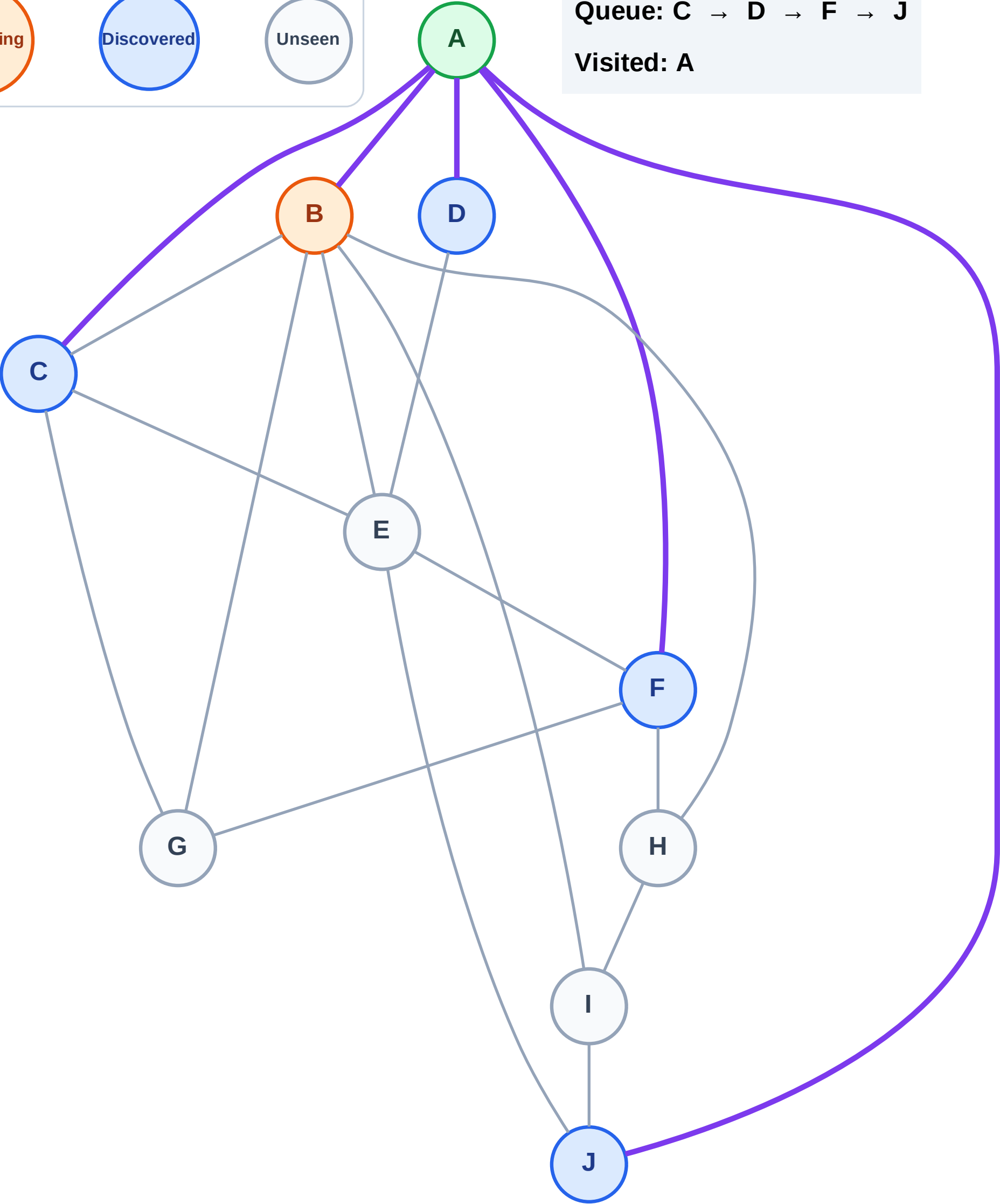
Processing

Discovered

Unseen

Queue: C → D → F → J

Visited: A



BFS Traversal

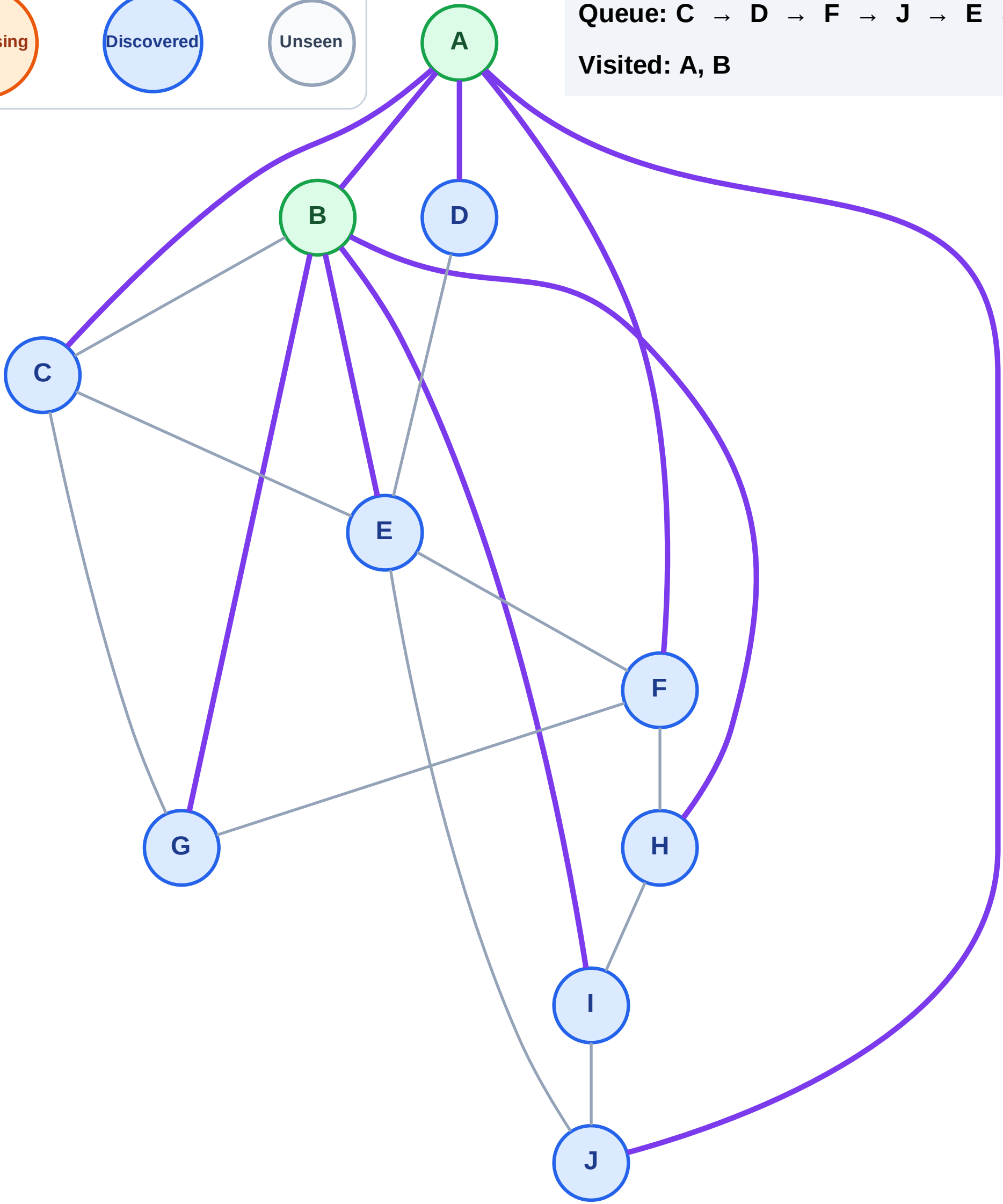
Step 5: Discover E, G, H, I from B

Legend



Queue: C → D → F → J → E → G → H → I

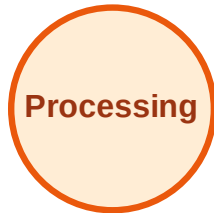
Visited: A, B



BFS Traversal

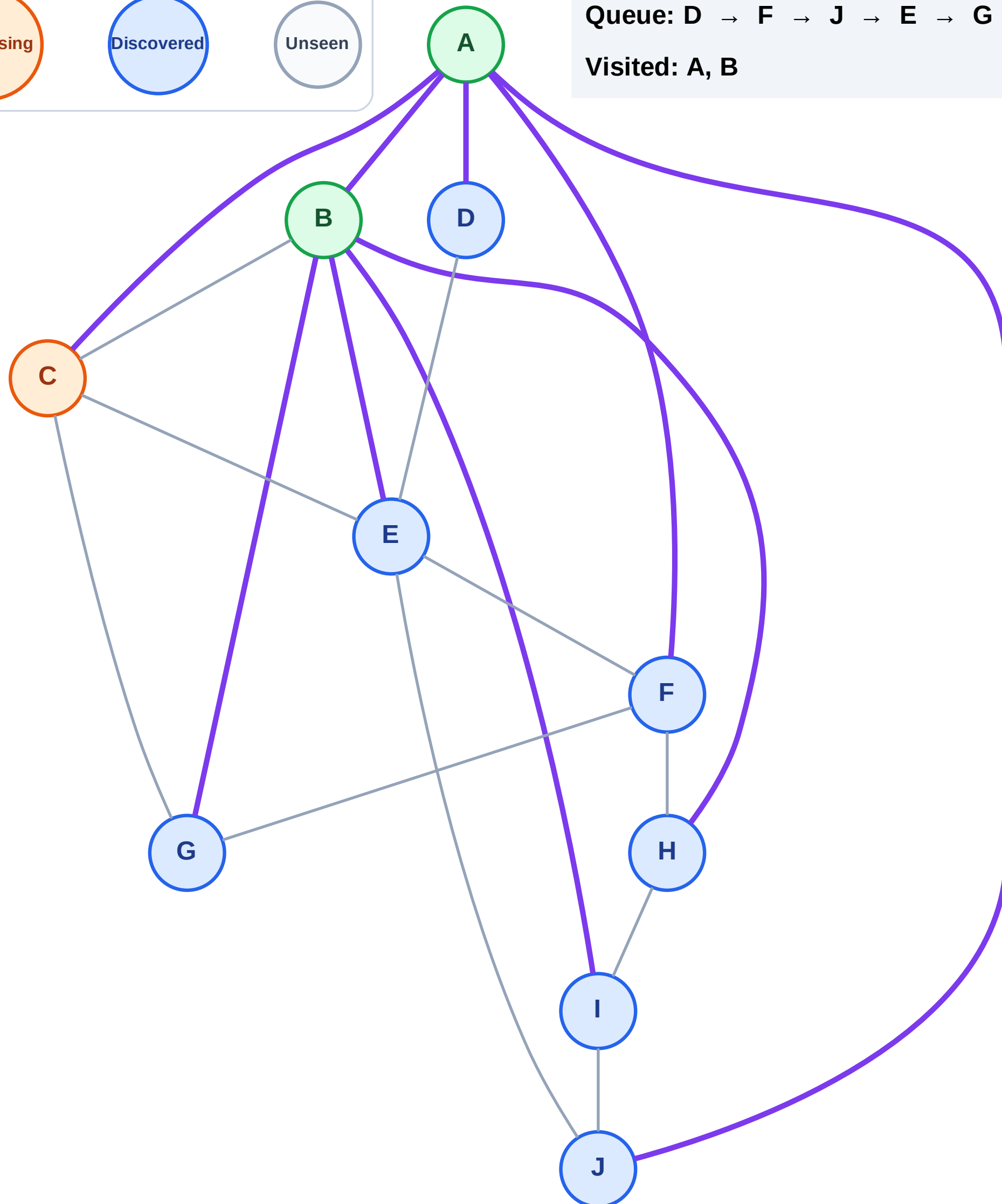
Step 6: Process node C

Legend



Queue: D → F → J → E → G → H → I

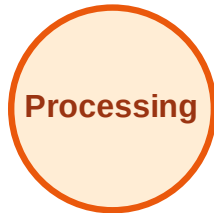
Visited: A, B



BFS Traversal

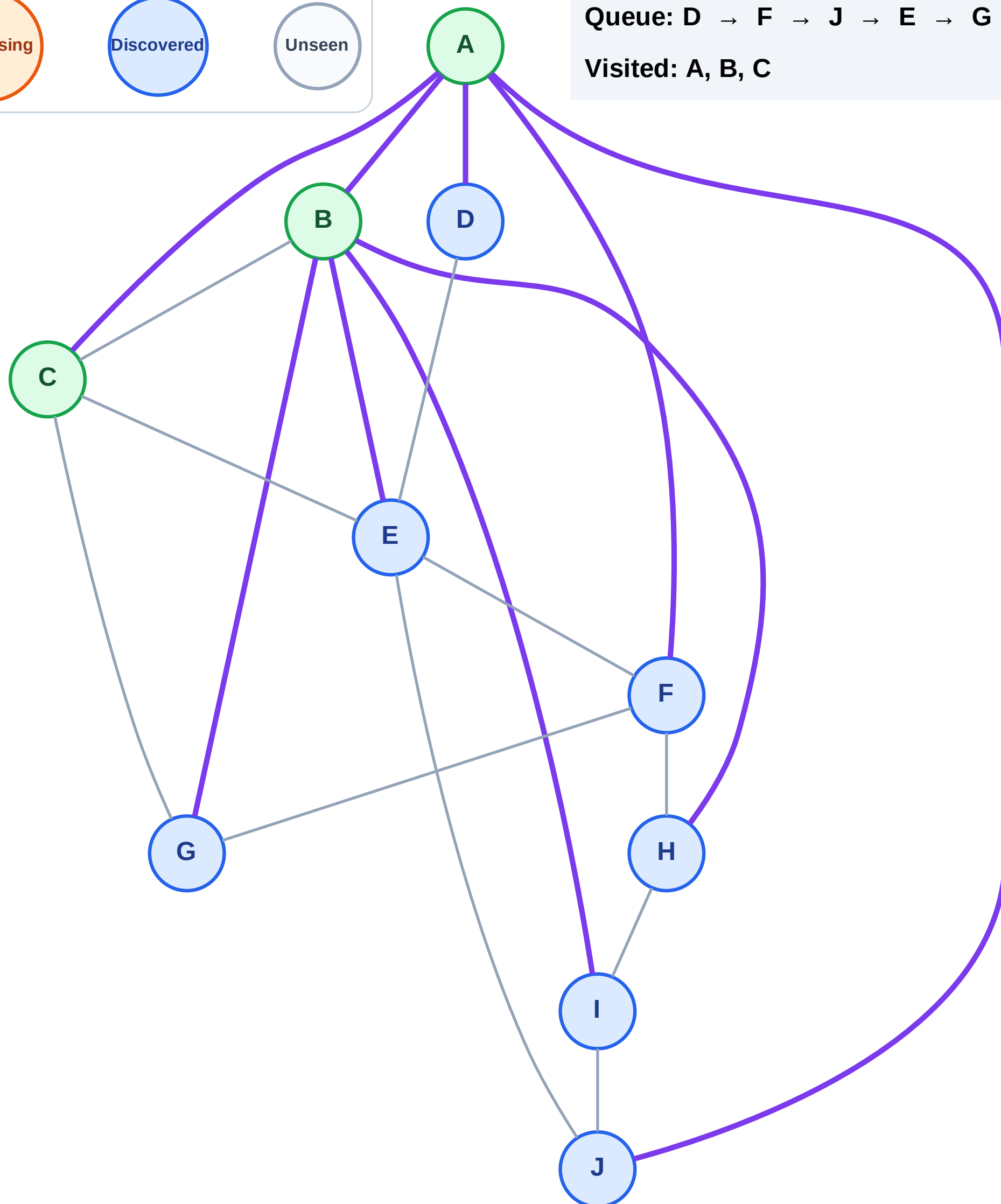
Step 7: No new neighbors from C

Legend



Queue: D → F → J → E → G → H → I

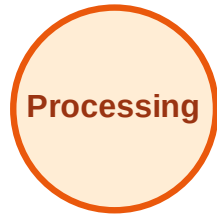
Visited: A, B, C



BFS Traversal

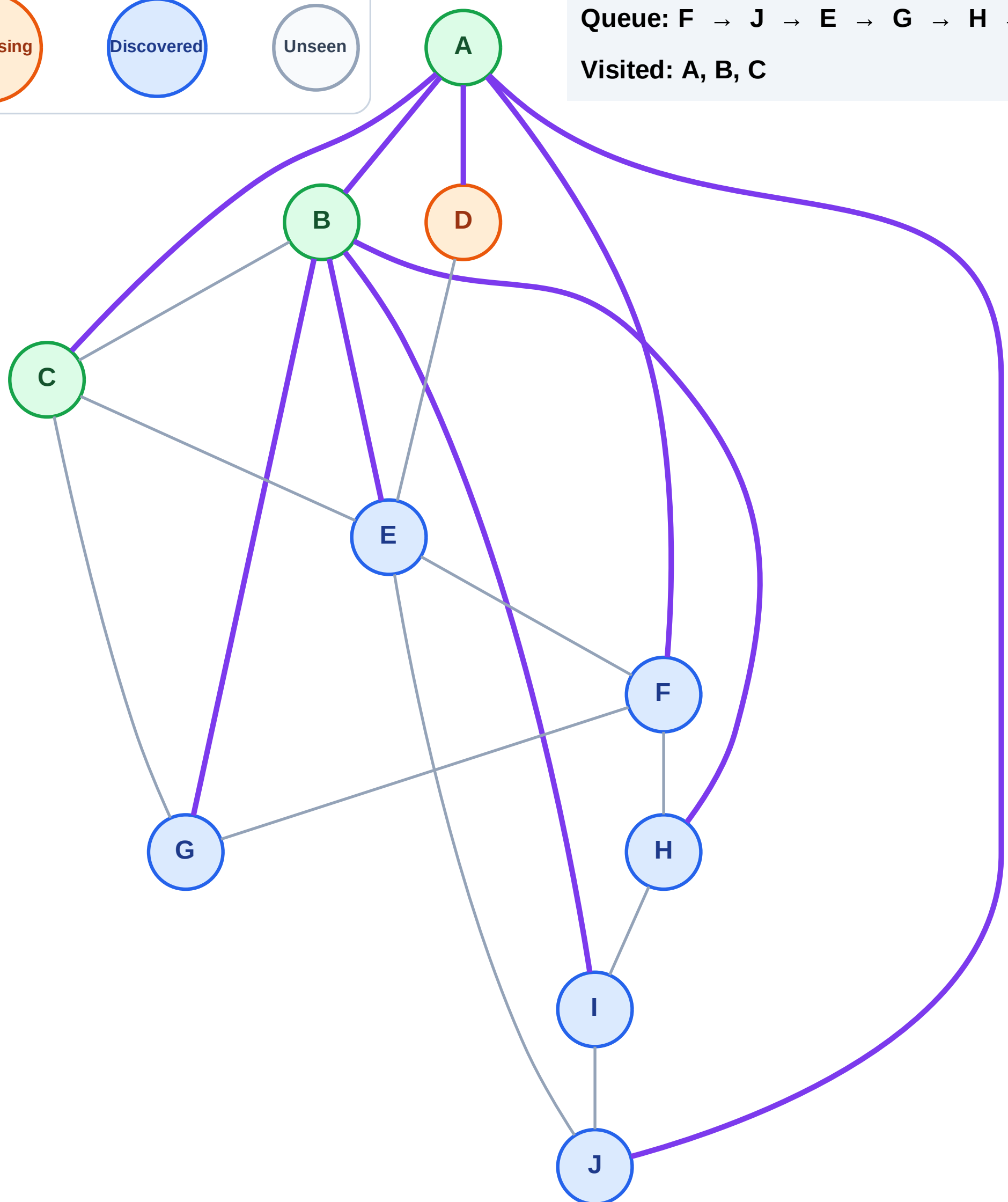
Step 8: Process node D

Legend



Queue: F → J → E → G → H → I

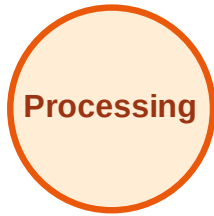
Visited: A, B, C



BFS Traversal

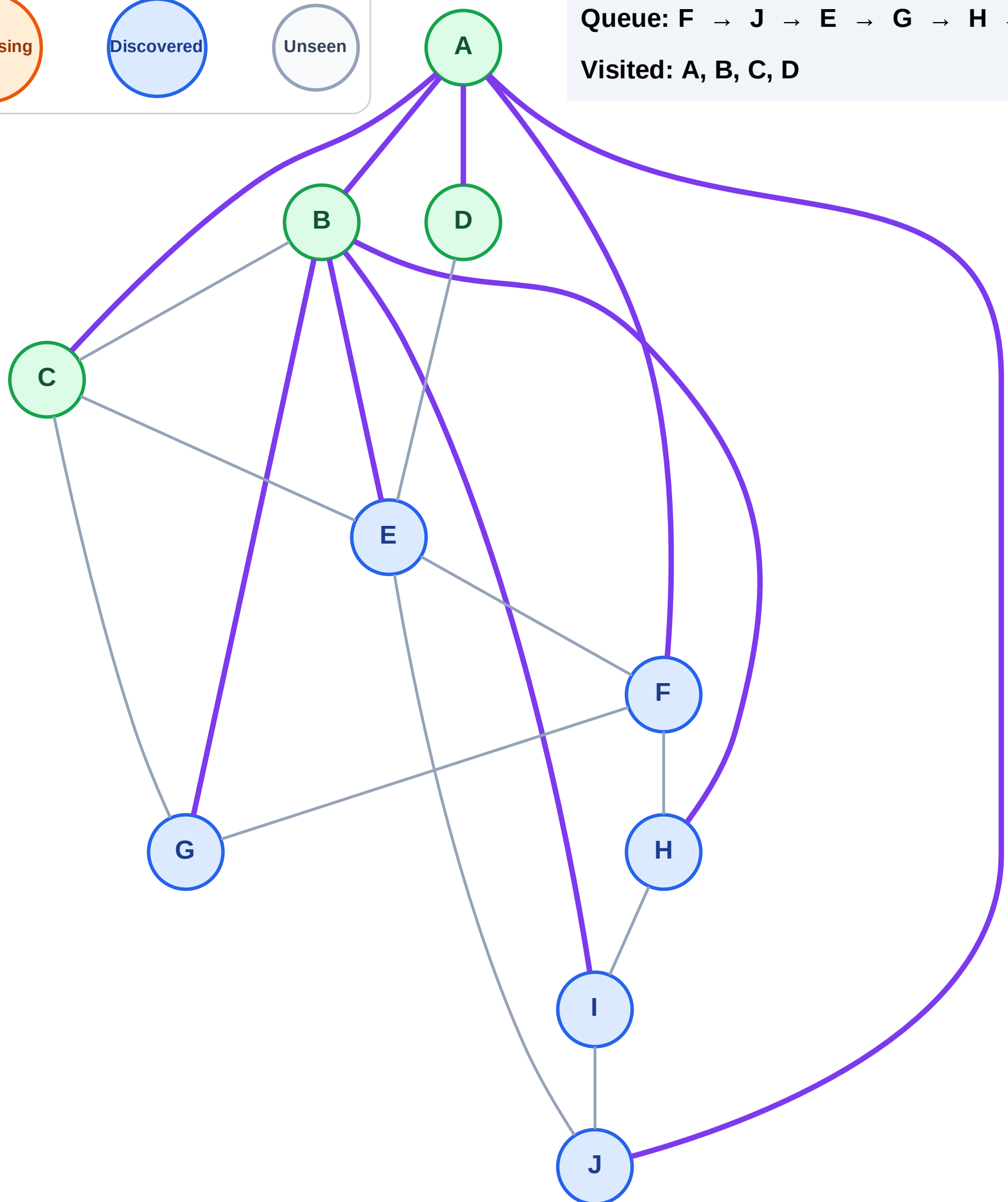
Step 9: No new neighbors from D

Legend



Queue: F → J → E → G → H → I

Visited: A, B, C, D



BFS Traversal

Step 10: Process node F

Legend

Complete

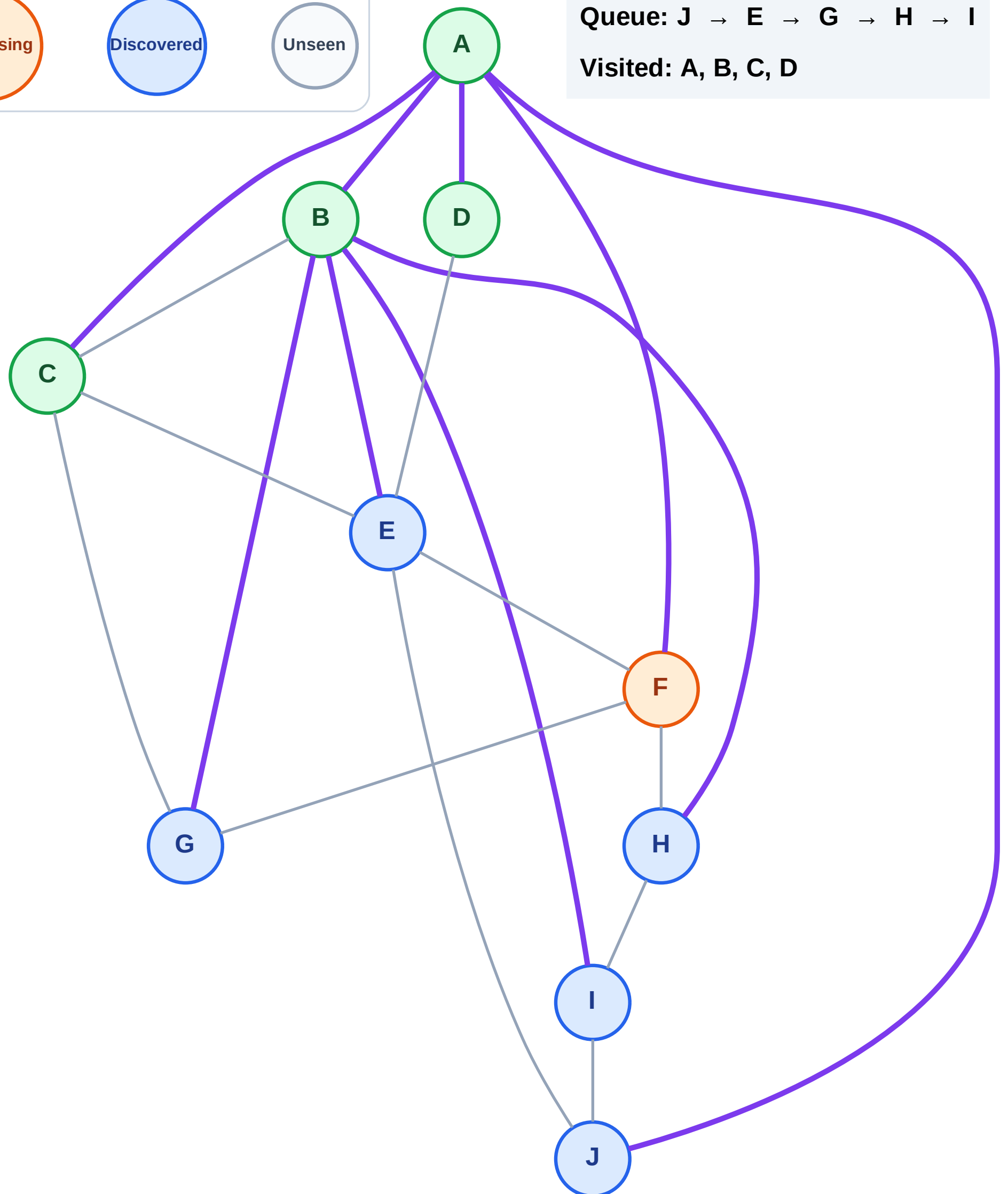
Processing

Discovered

Unseen

Queue: J → E → G → H → I

Visited: A, B, C, D



BFS Traversal

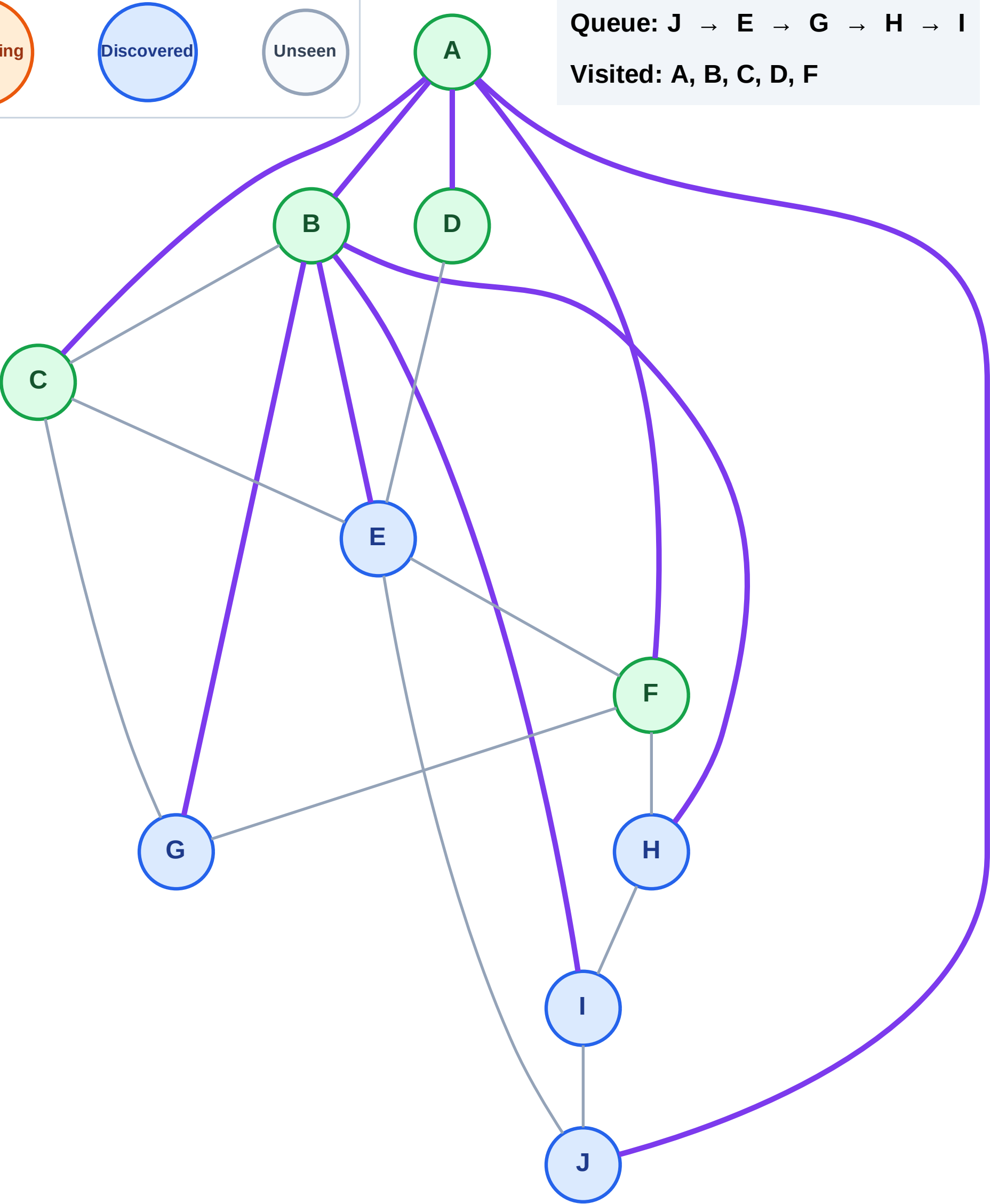
Step 11: No new neighbors from F

Legend



Queue: J → E → G → H → I

Visited: A, B, C, D, F



BFS Traversal

Step 12: Process node J

Legend

Complete

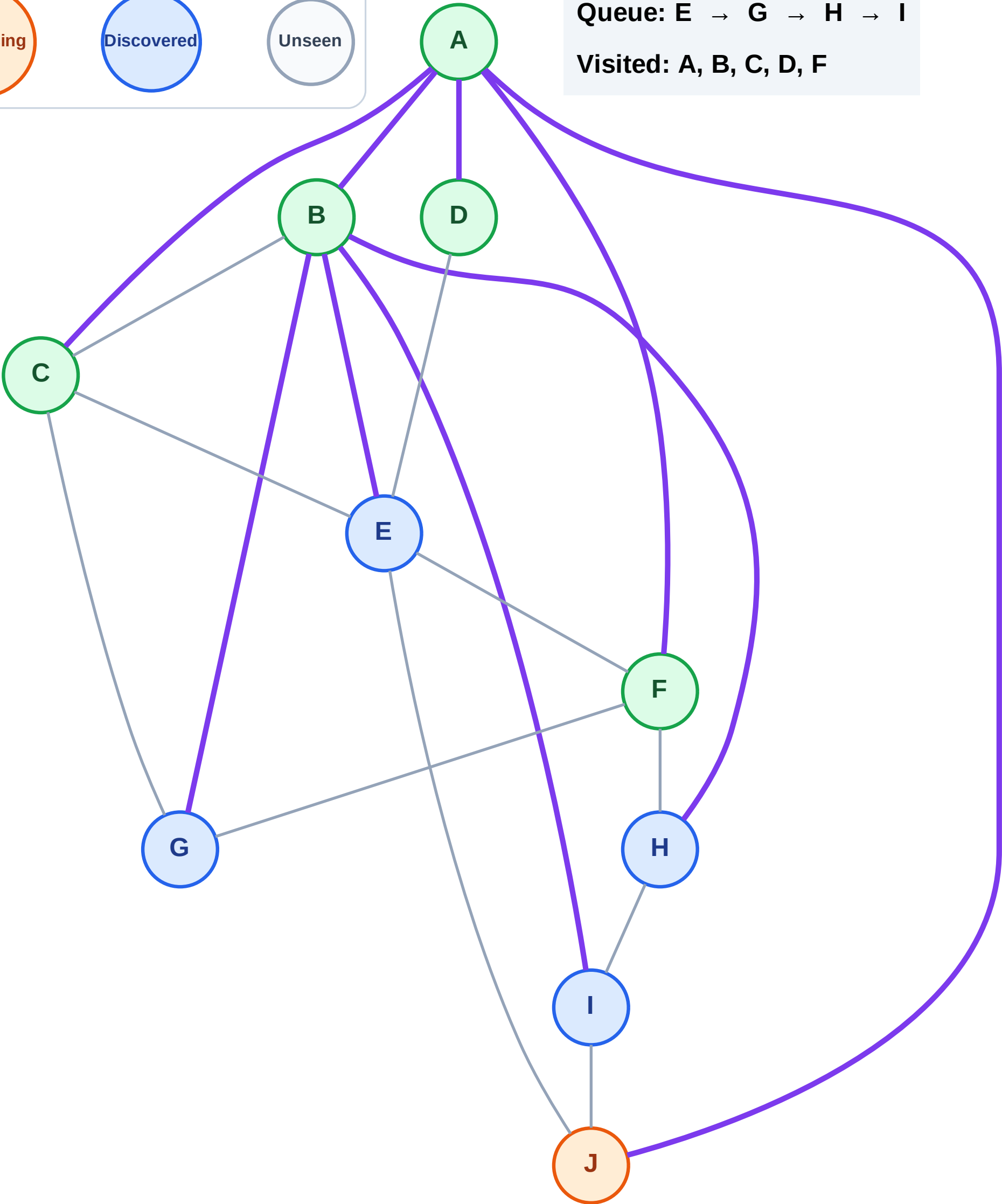
Processing

Discovered

Unseen

Queue: E → G → H → I

Visited: A, B, C, D, F



BFS Traversal

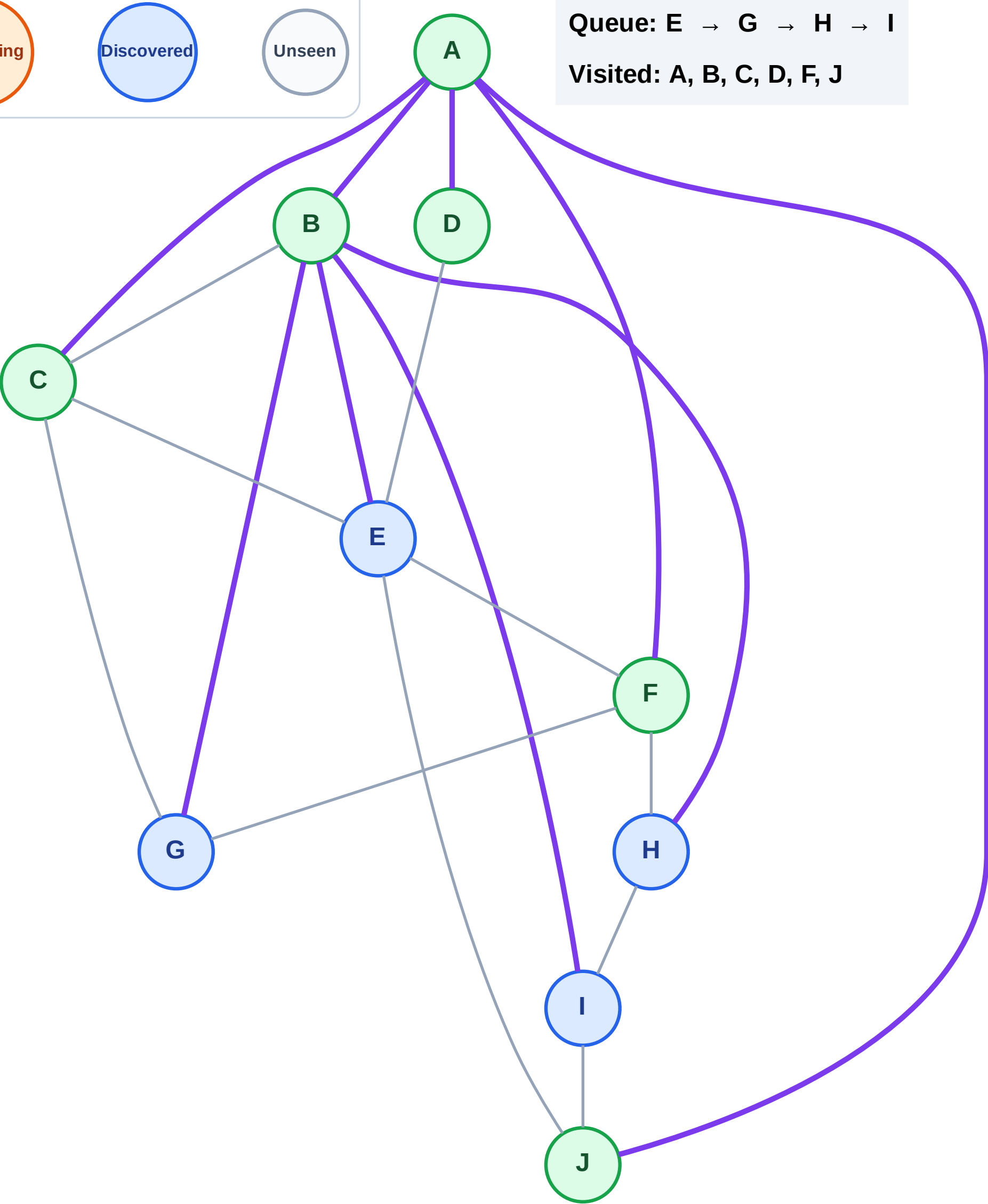
Step 13: No new neighbors from J

Legend



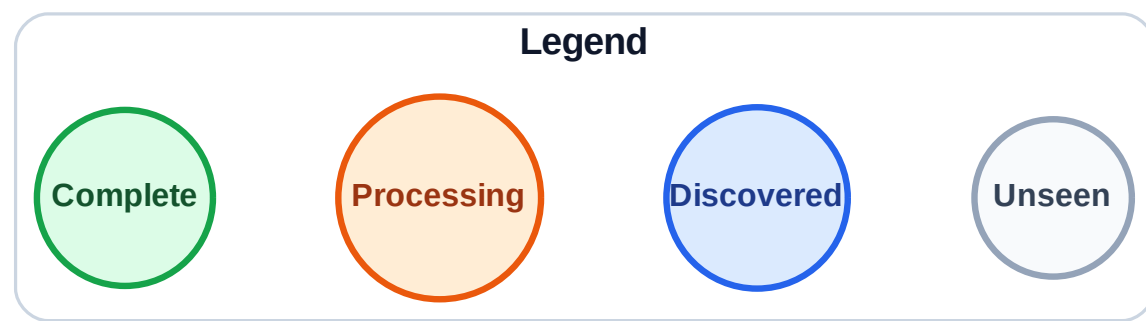
Queue: E → G → H → I

Visited: A, B, C, D, F, J

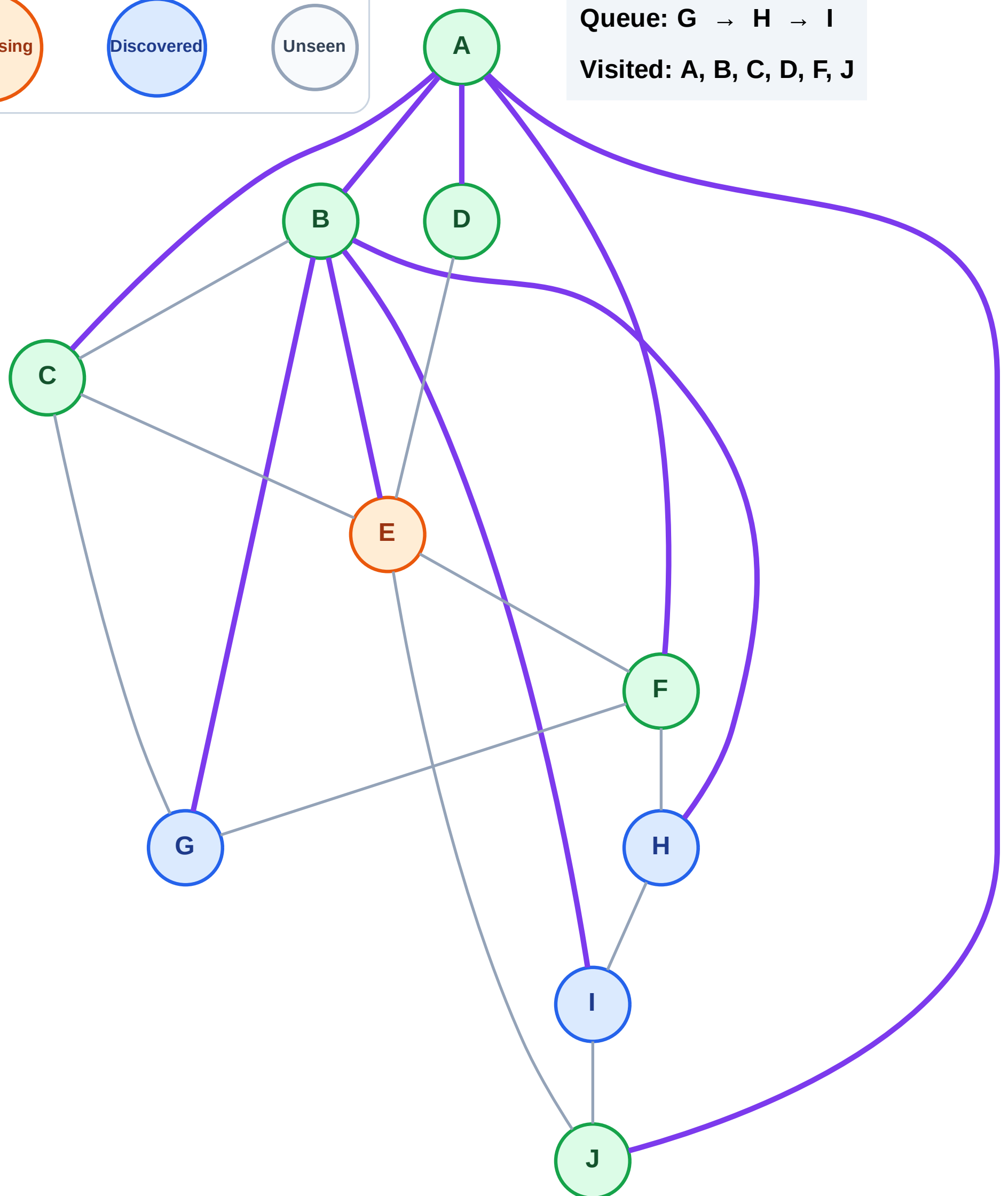


BFS Traversal

Step 14: Process node E



Queue: G → H → I
Visited: A, B, C, D, F, J



BFS Traversal

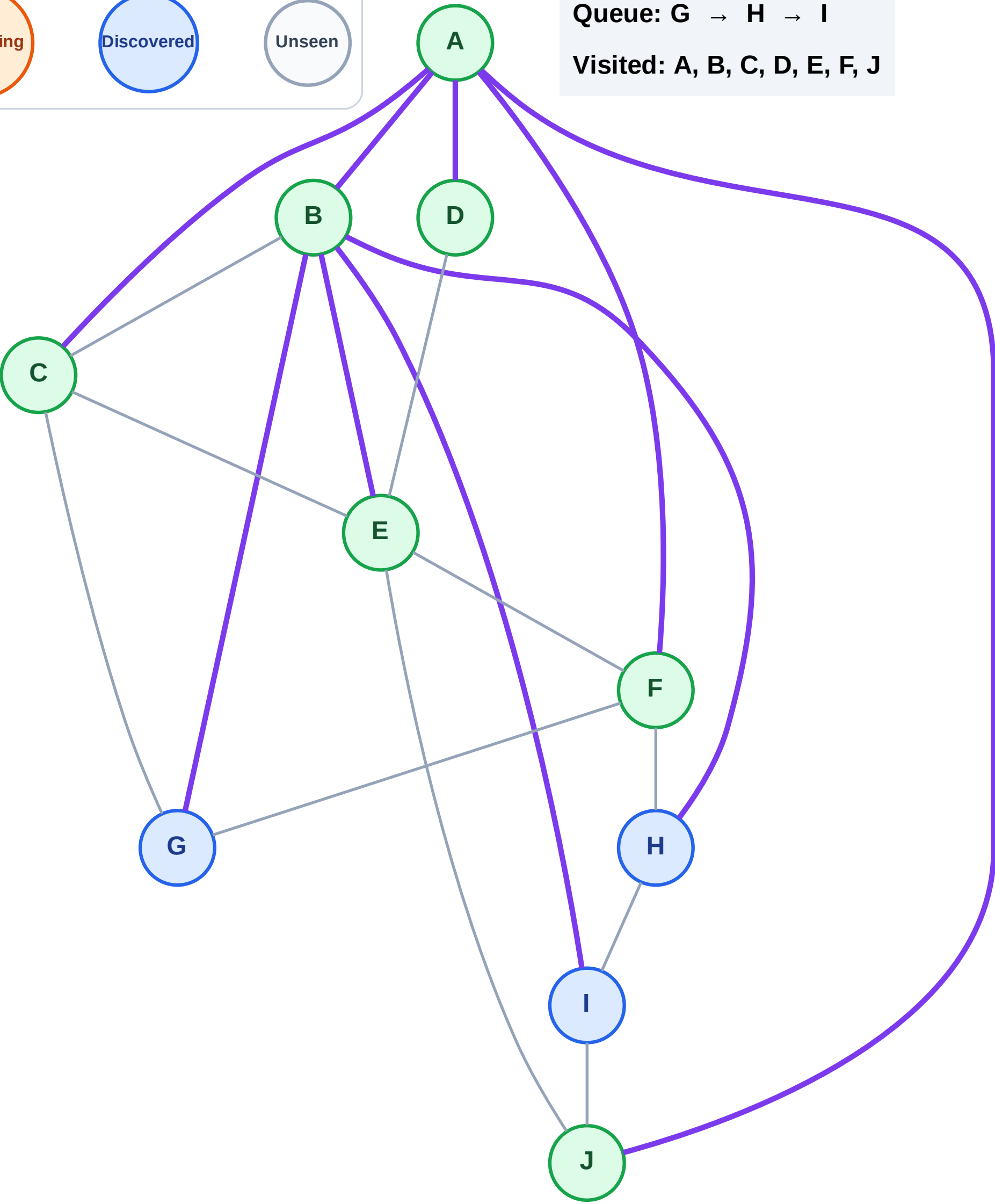
Step 15: No new neighbors from E

Legend



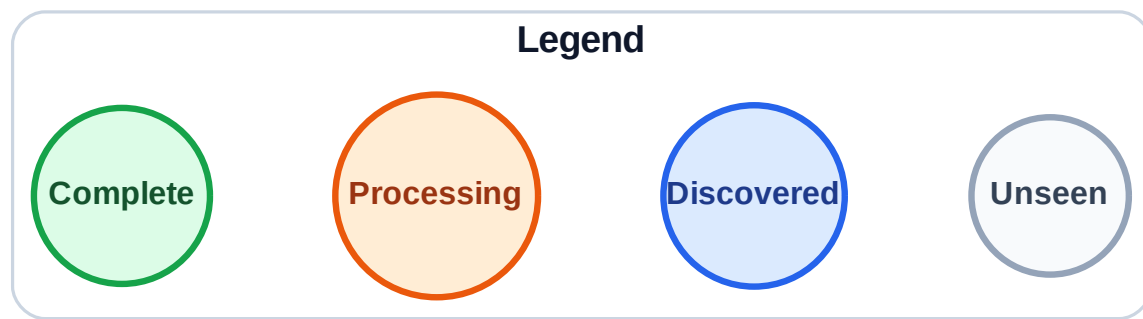
Queue: G → H → I

Visited: A, B, C, D, E, F, J

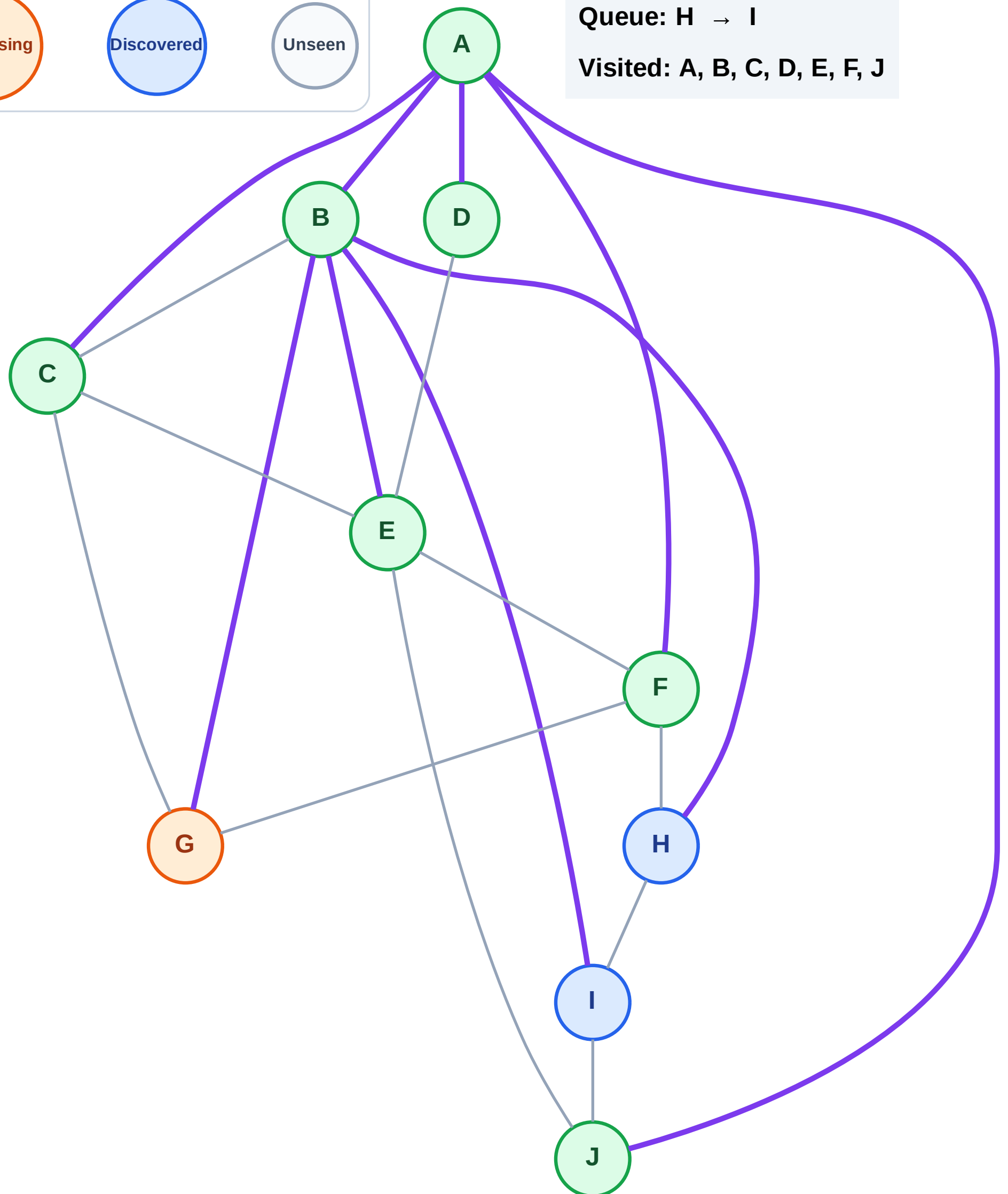


BFS Traversal

Step 16: Process node G



Queue: H → I
Visited: A, B, C, D, E, F, J



BFS Traversal

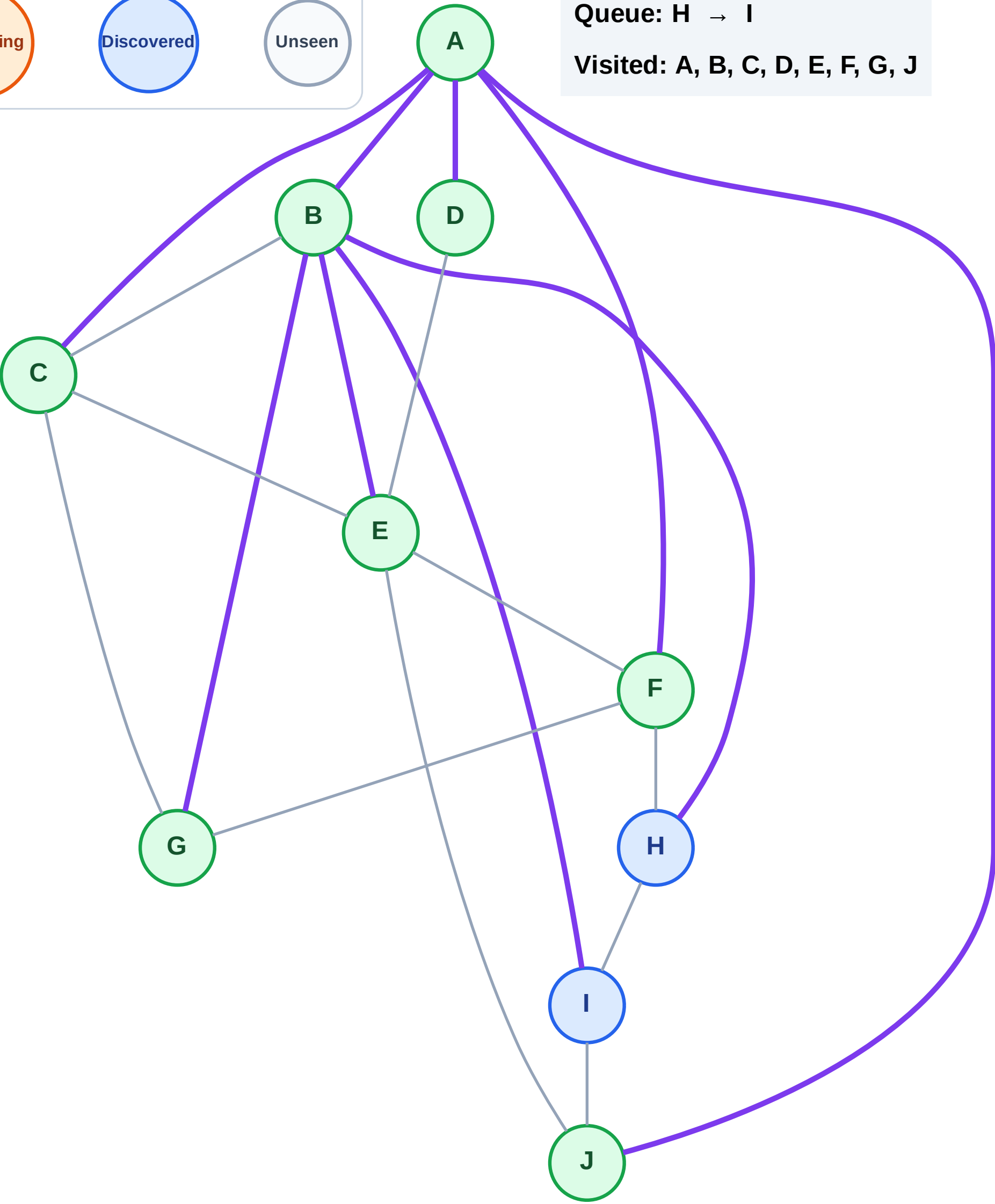
Step 17: No new neighbors from G

Legend



Queue: H → I

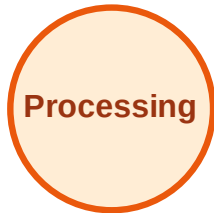
Visited: A, B, C, D, E, F, G, J



BFS Traversal

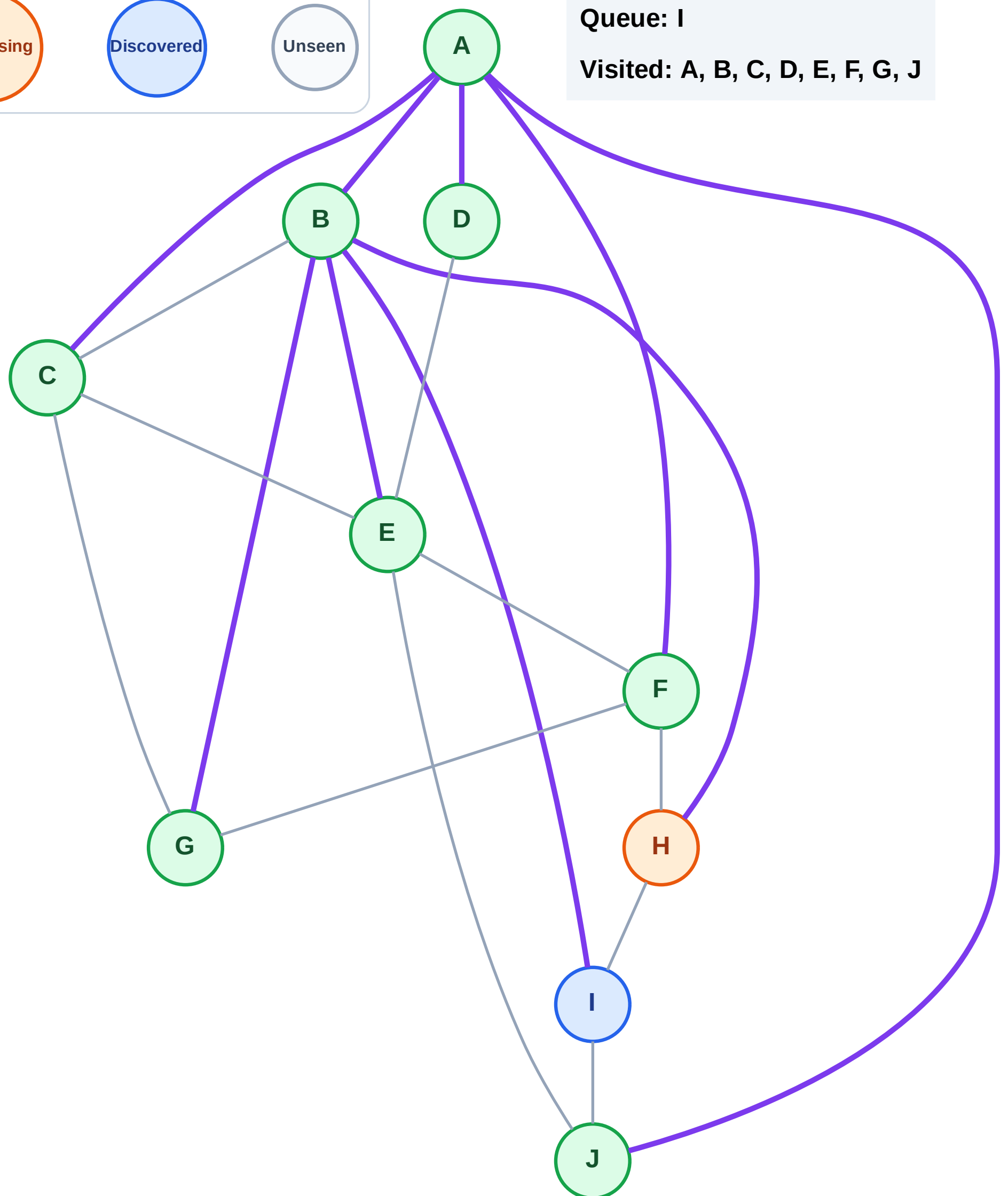
Step 18: Process node H

Legend



Queue: I

Visited: A, B, C, D, E, F, G, J



BFS Traversal

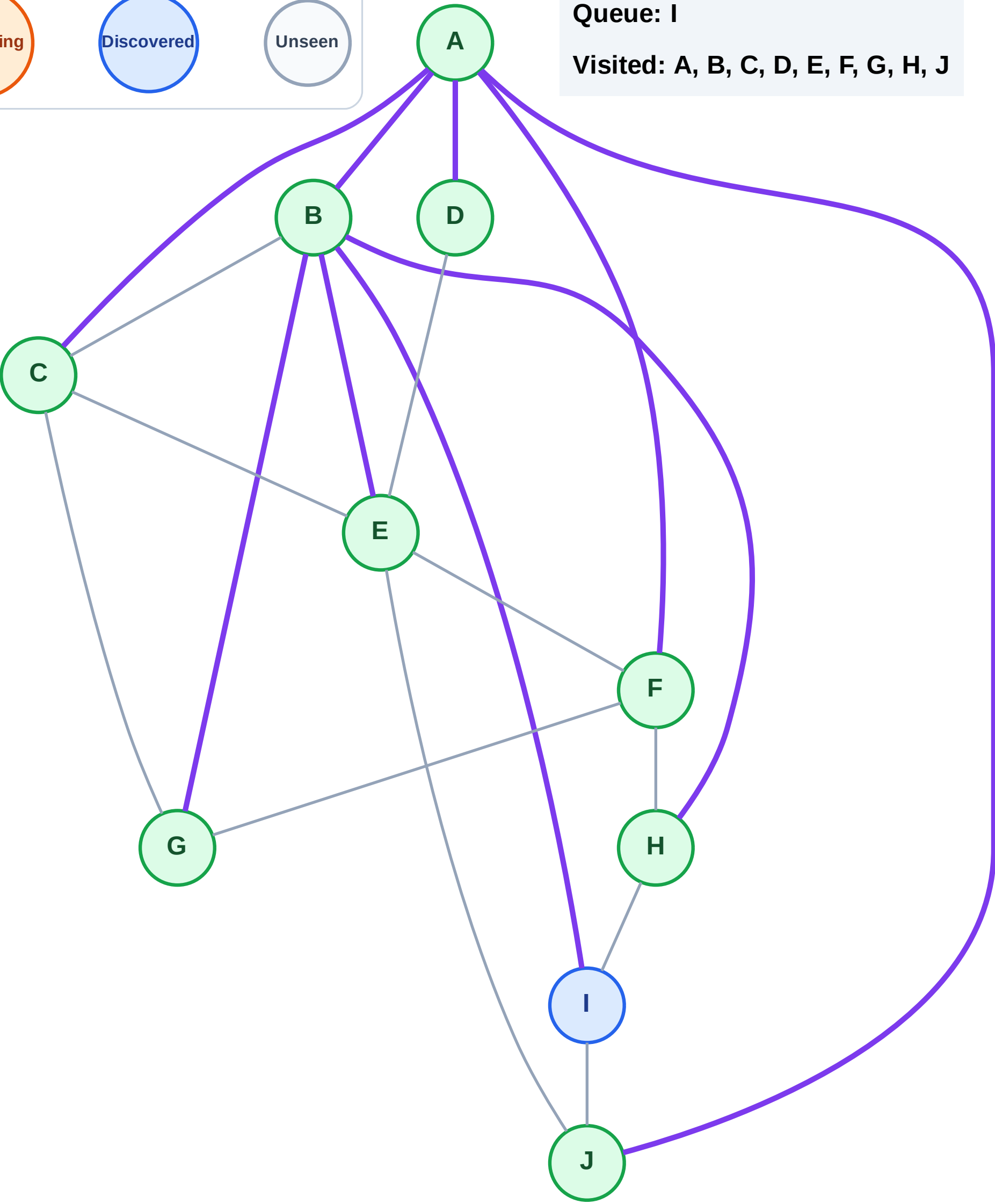
Step 19: No new neighbors from H

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

Step 20: Process node I

Legend

Complete

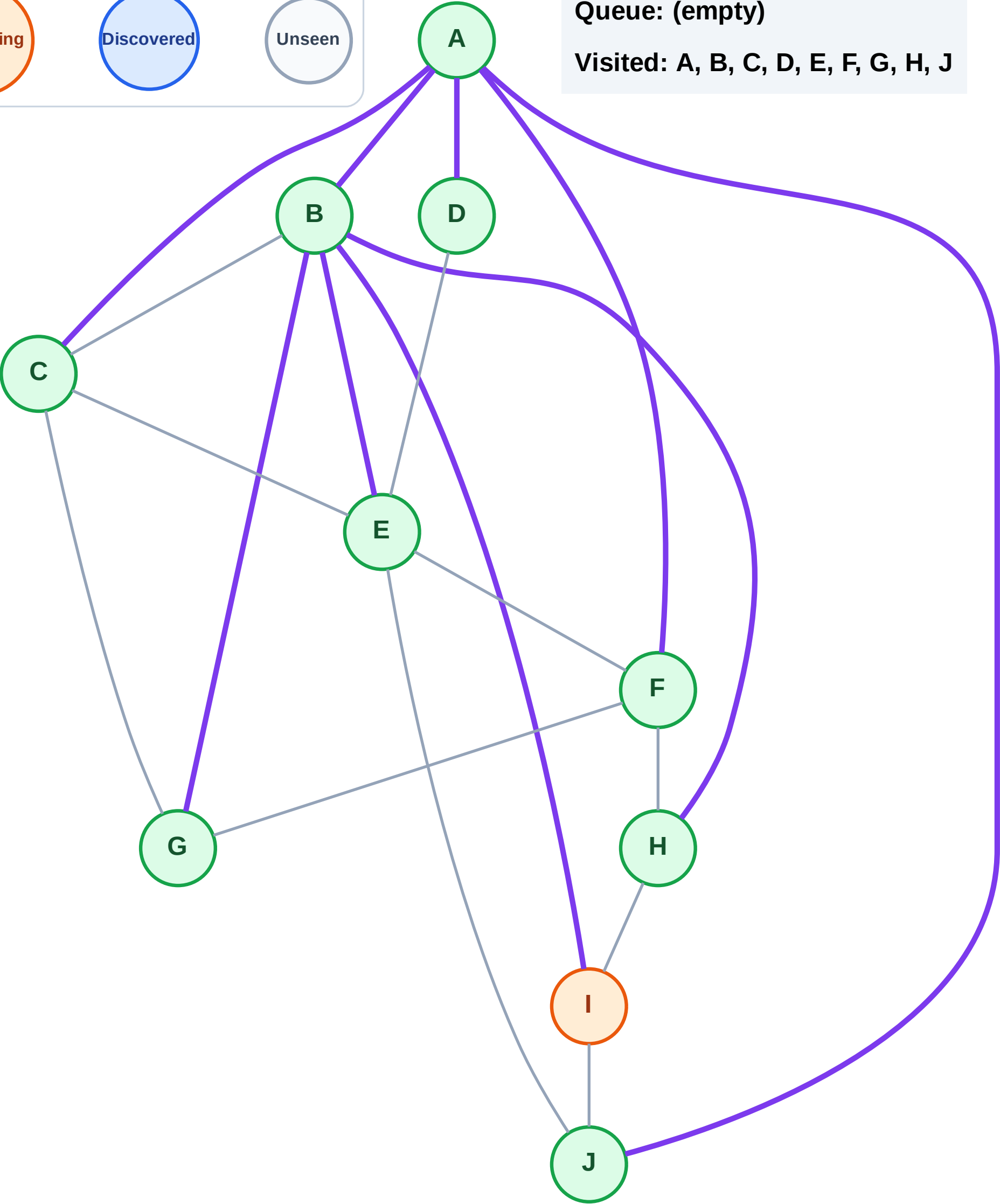
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

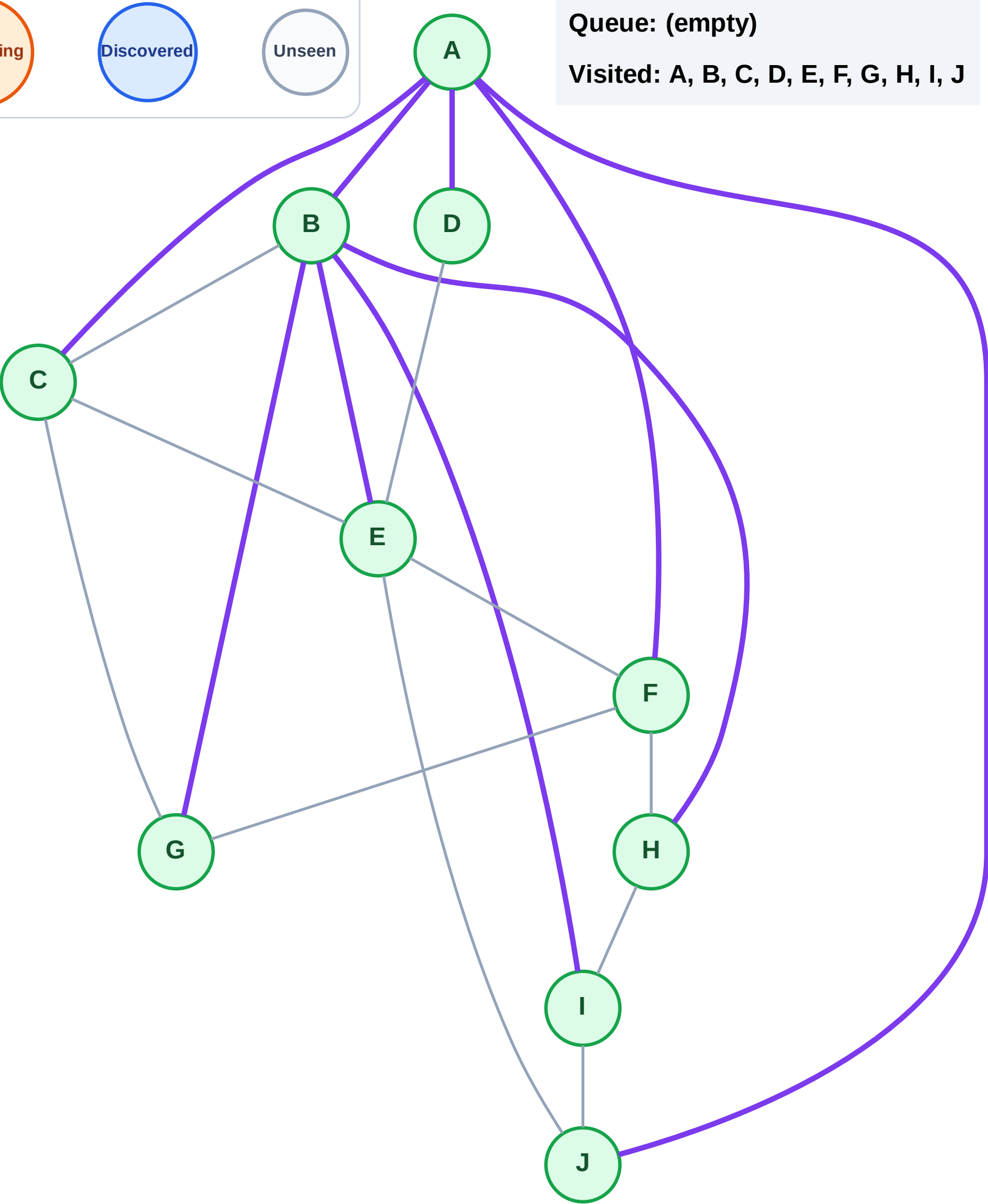
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

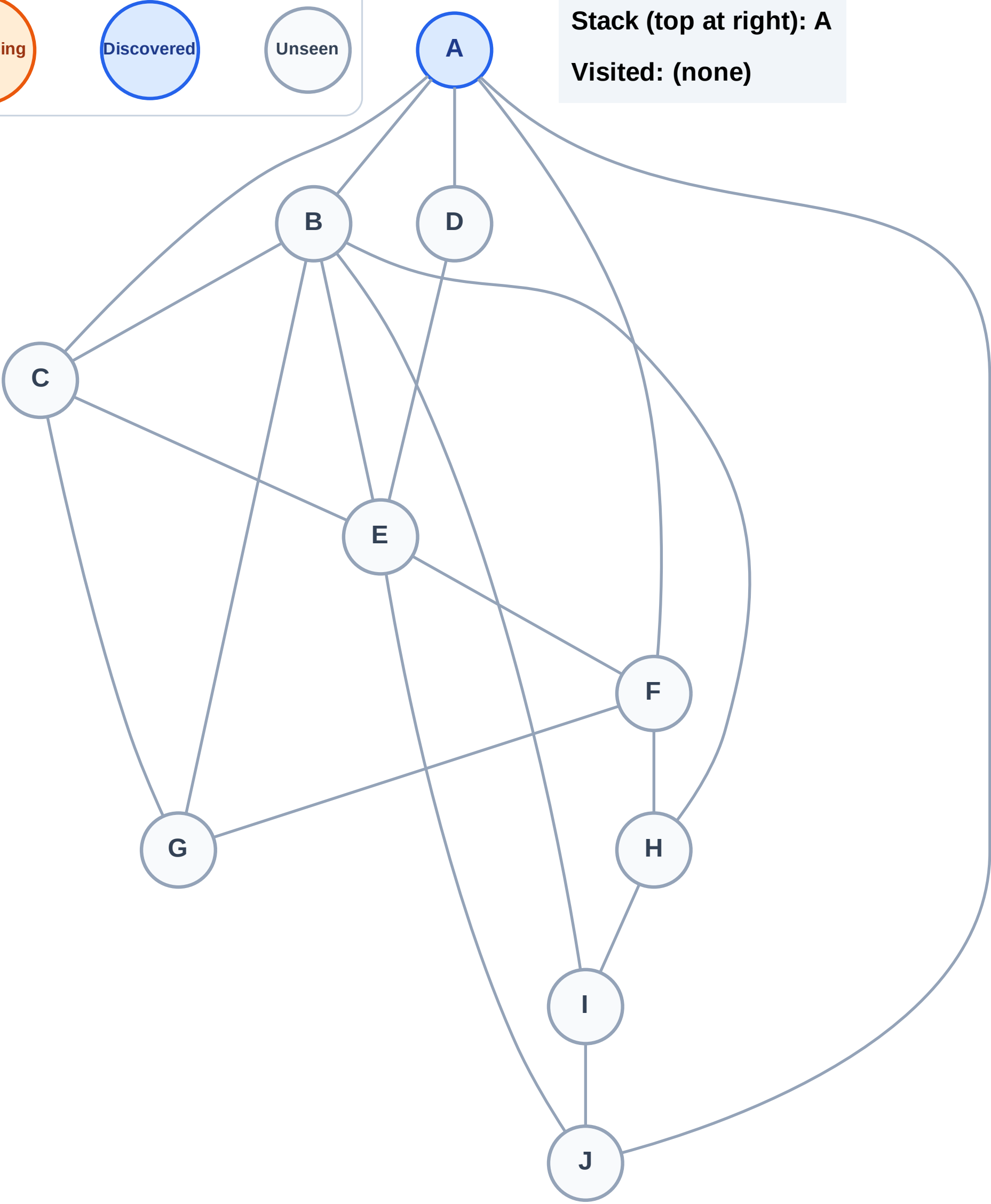
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

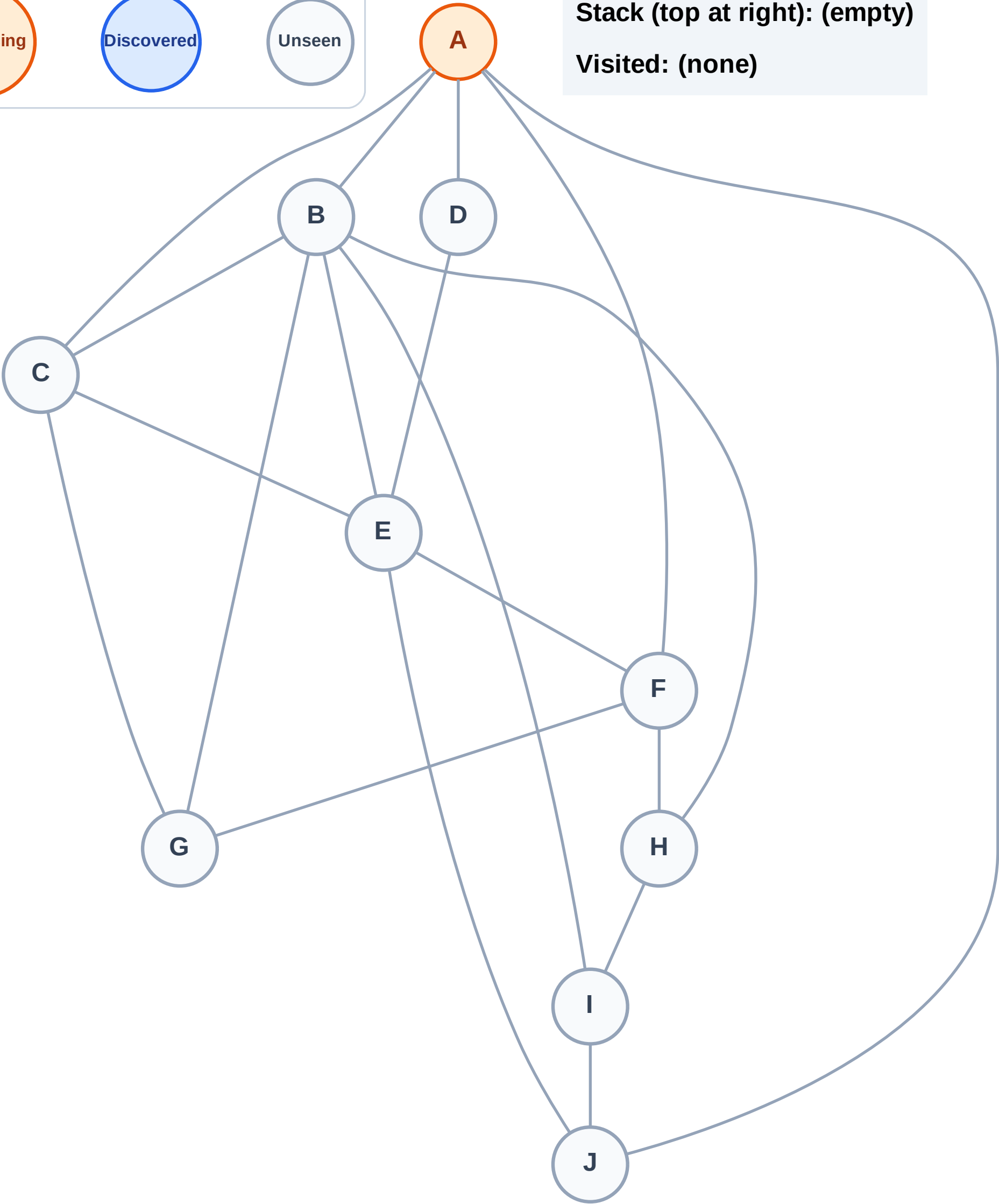
Complete

Processing

Discovered

Unseen

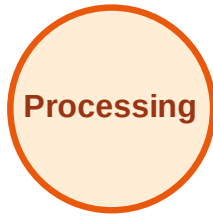
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

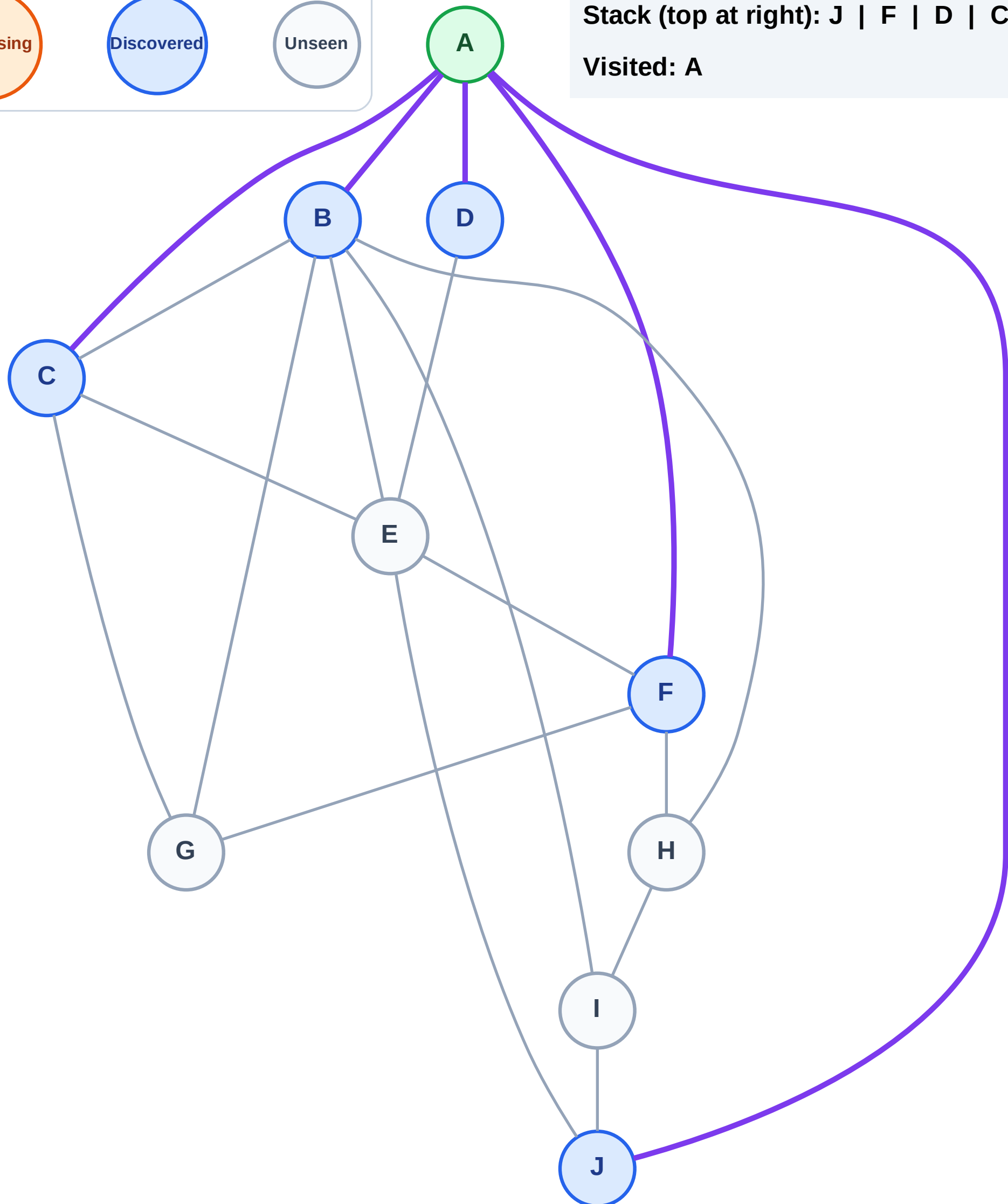
Step 3: Discover J, F, D, C, B from A

Legend



Stack (top at right): J | F | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

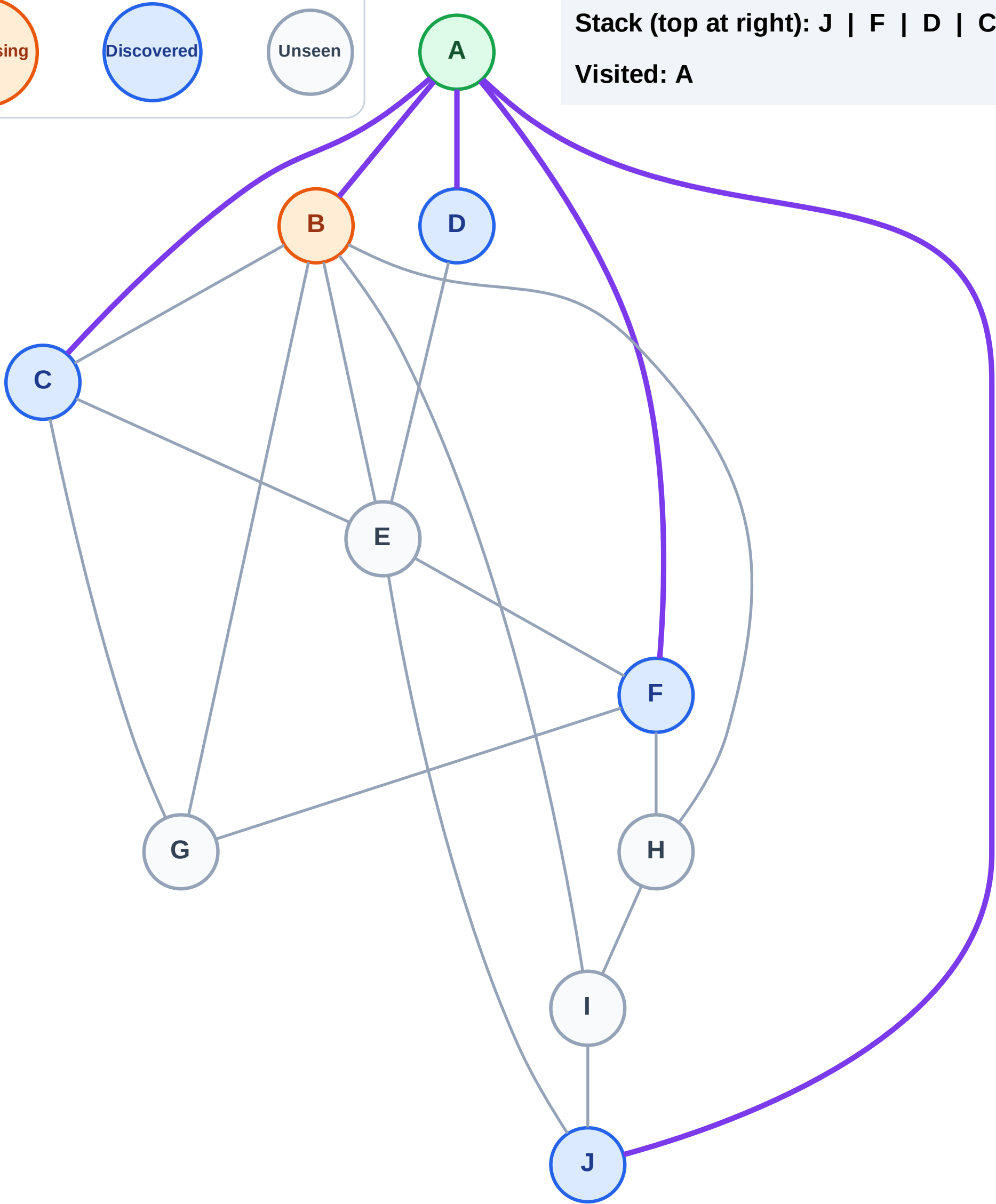
Processing

Discovered

Unseen

Stack (top at right): J | F | D | C

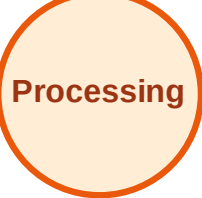
Visited: A



DFS Traversal

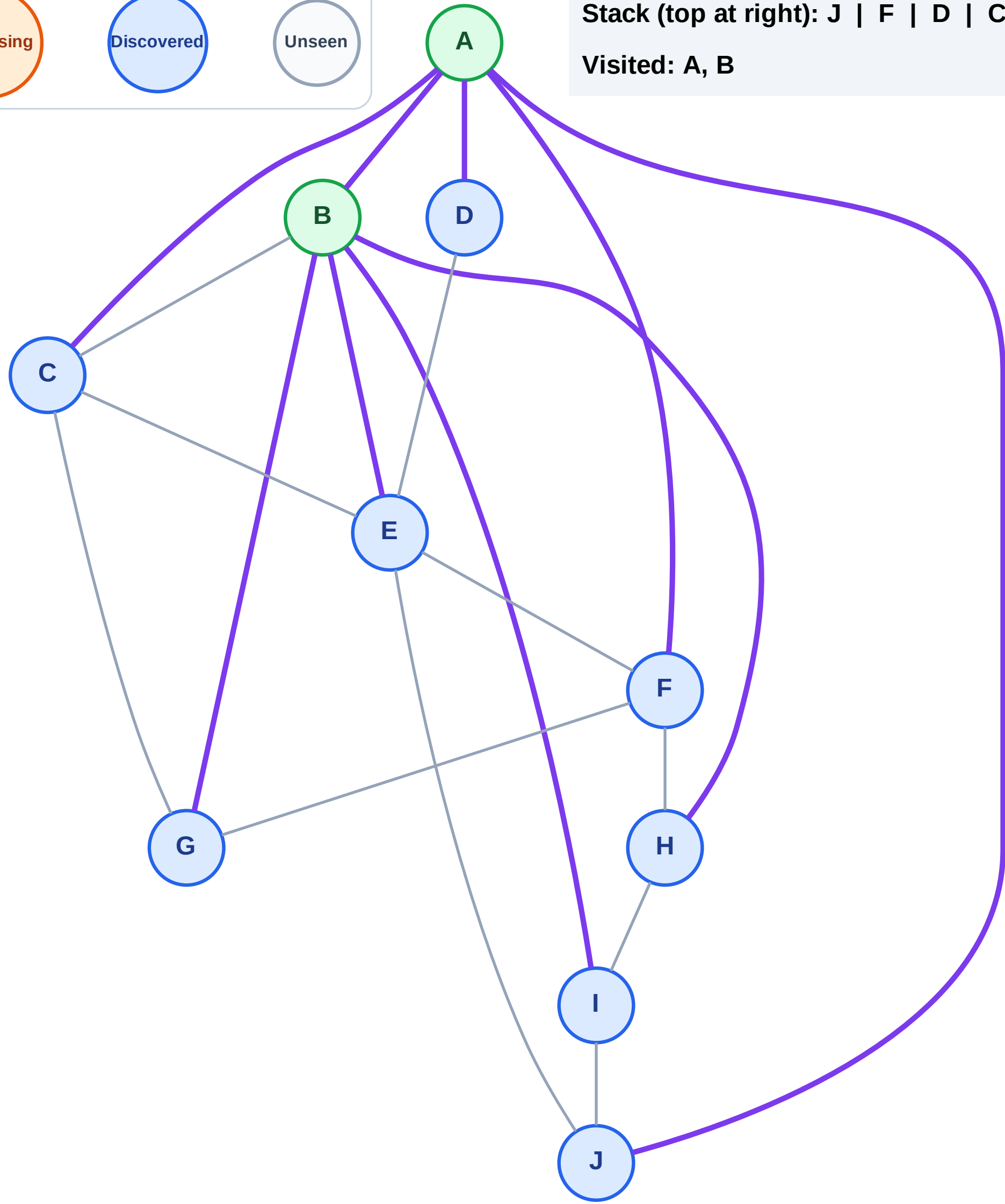
Step 5: Discover I, H, G, E from B

Legend



Stack (top at right): J | F | D | C | I | H | G | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

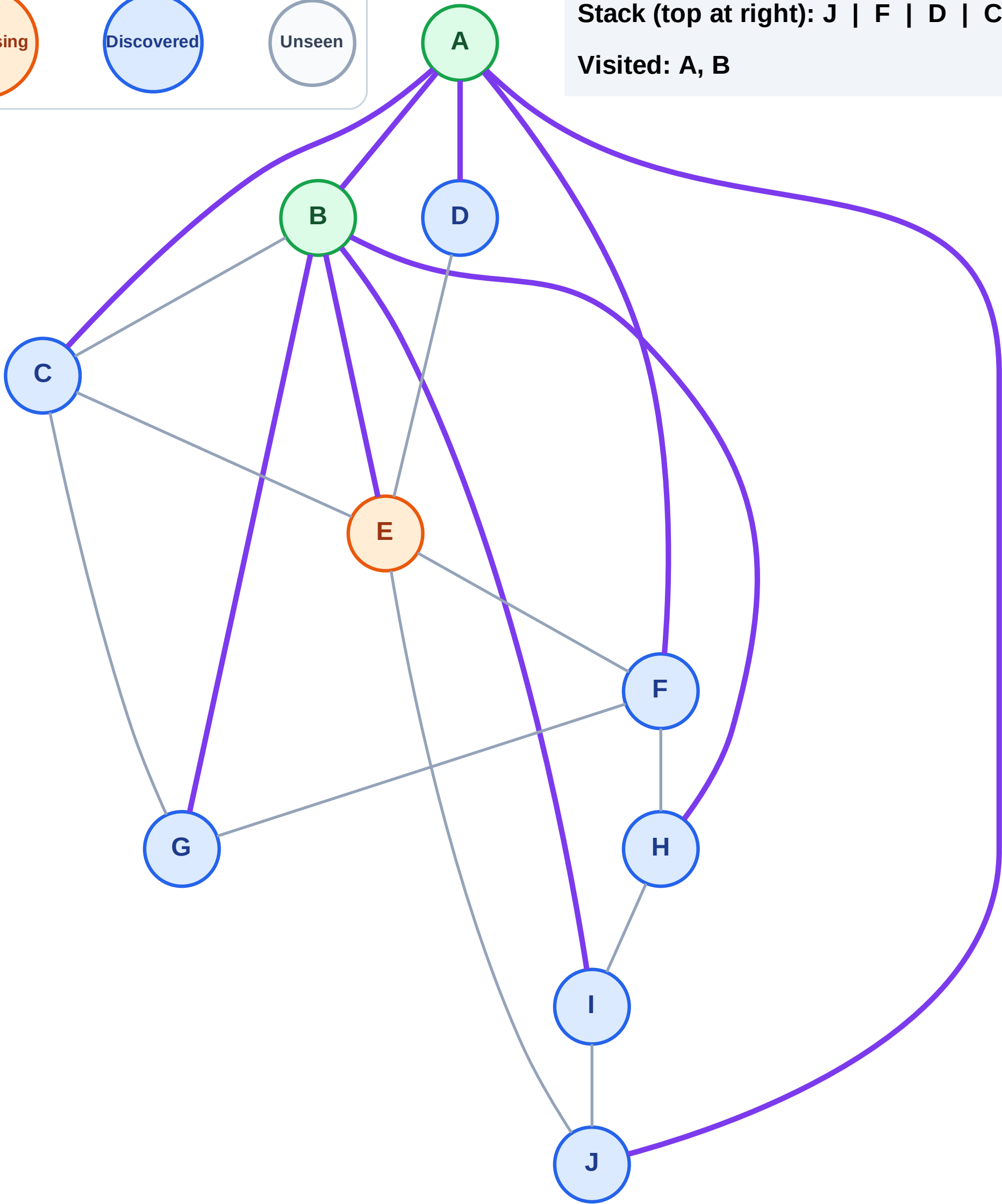
Processing

Discovered

Unseen

Stack (top at right): J | F | D | C | I | H | G

Visited: A, B



DFS Traversal

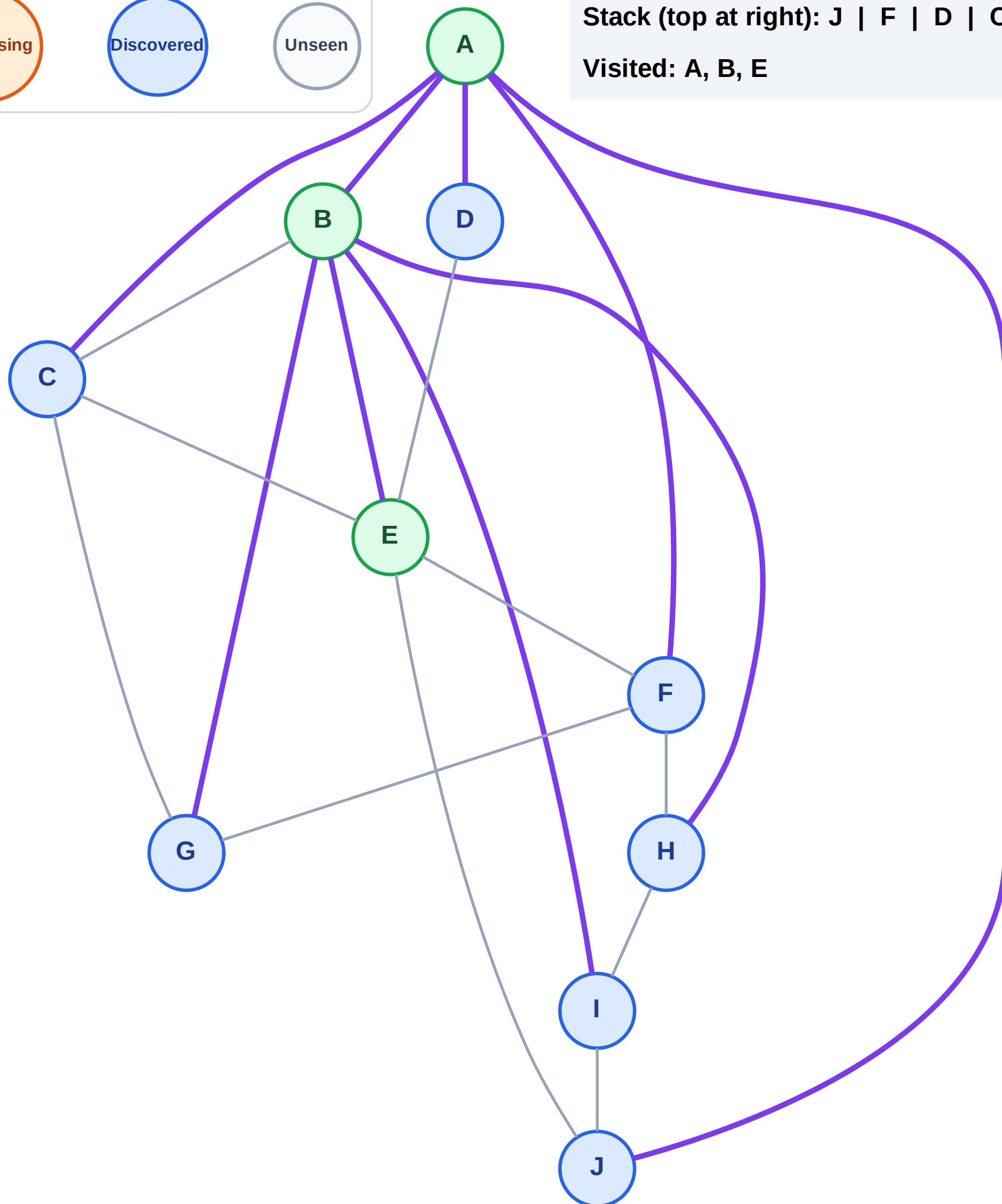
Step 7: No new neighbors from E

Legend



Stack (top at right): J | F | D | C | I | H | G

Visited: A, B, E



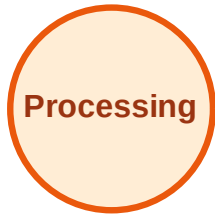
DFS Traversal

Step 8: Process node G

Legend



Complete



Processing



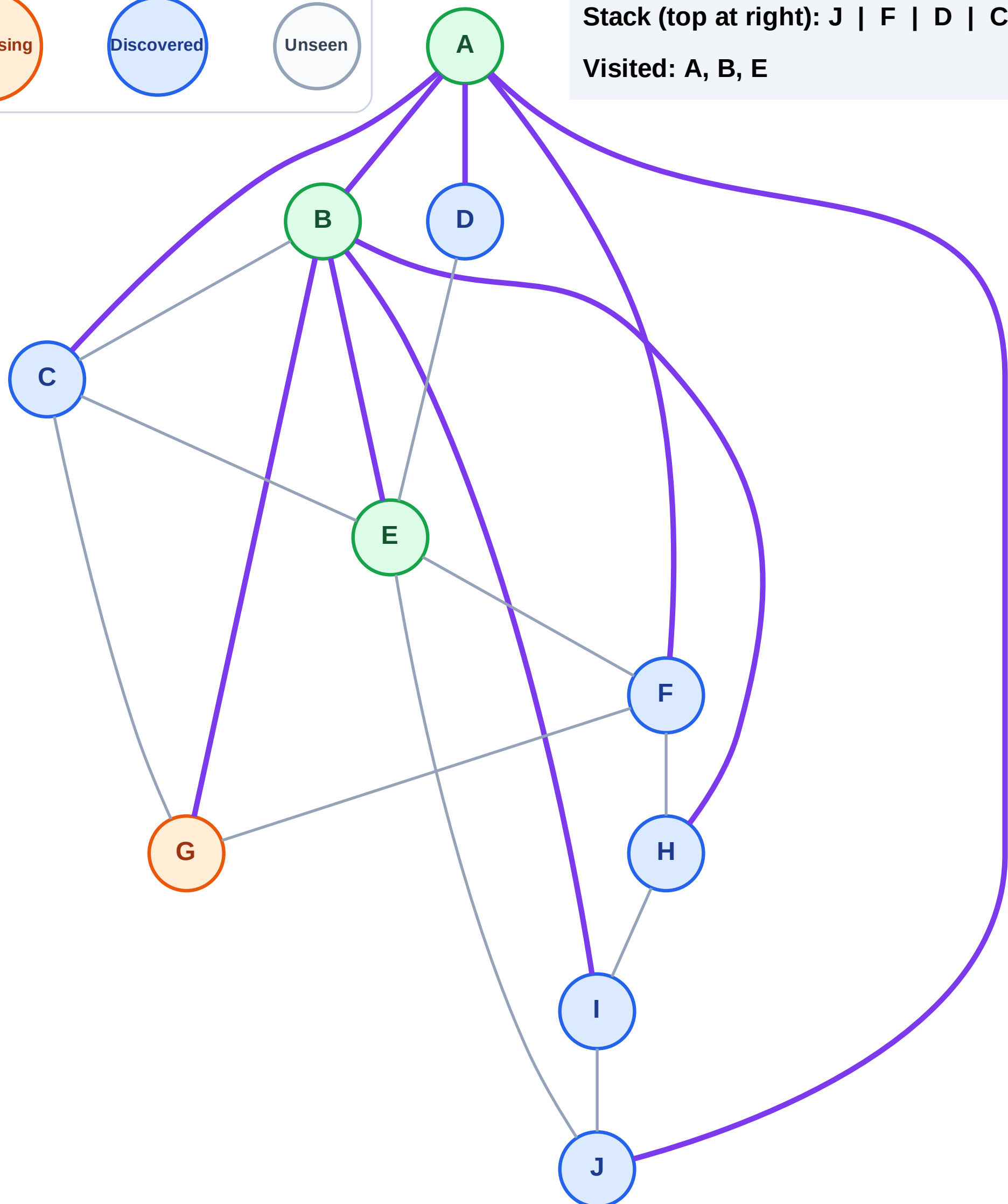
Discovered



Unseen

Stack (top at right): J | F | D | C | I | H

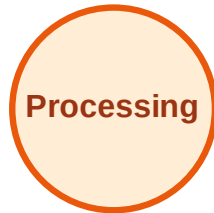
Visited: A, B, E



DFS Traversal

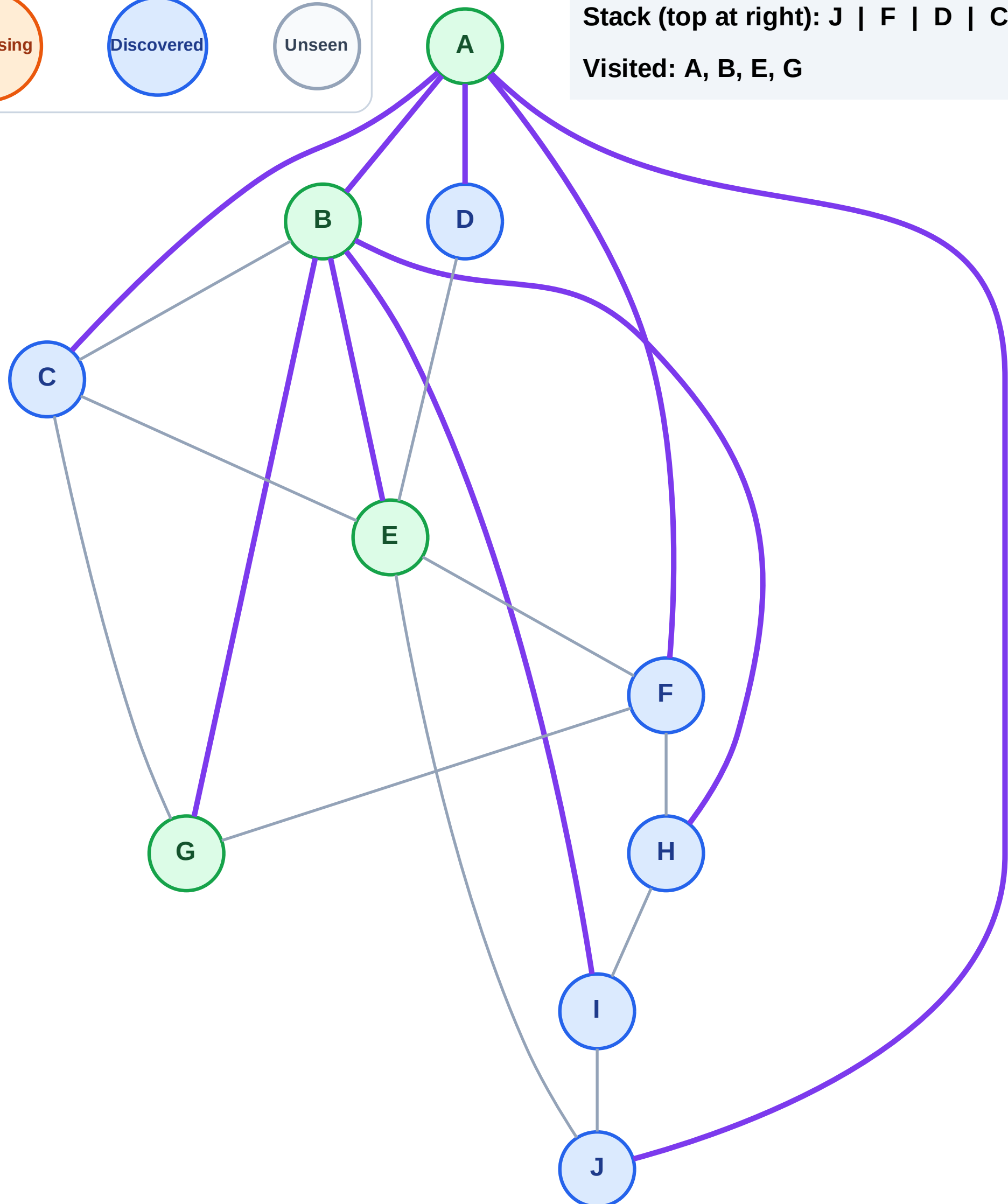
Step 9: No new neighbors from G

Legend



Stack (top at right): J | F | D | C | I | H

Visited: A, B, E, G



DFS Traversal

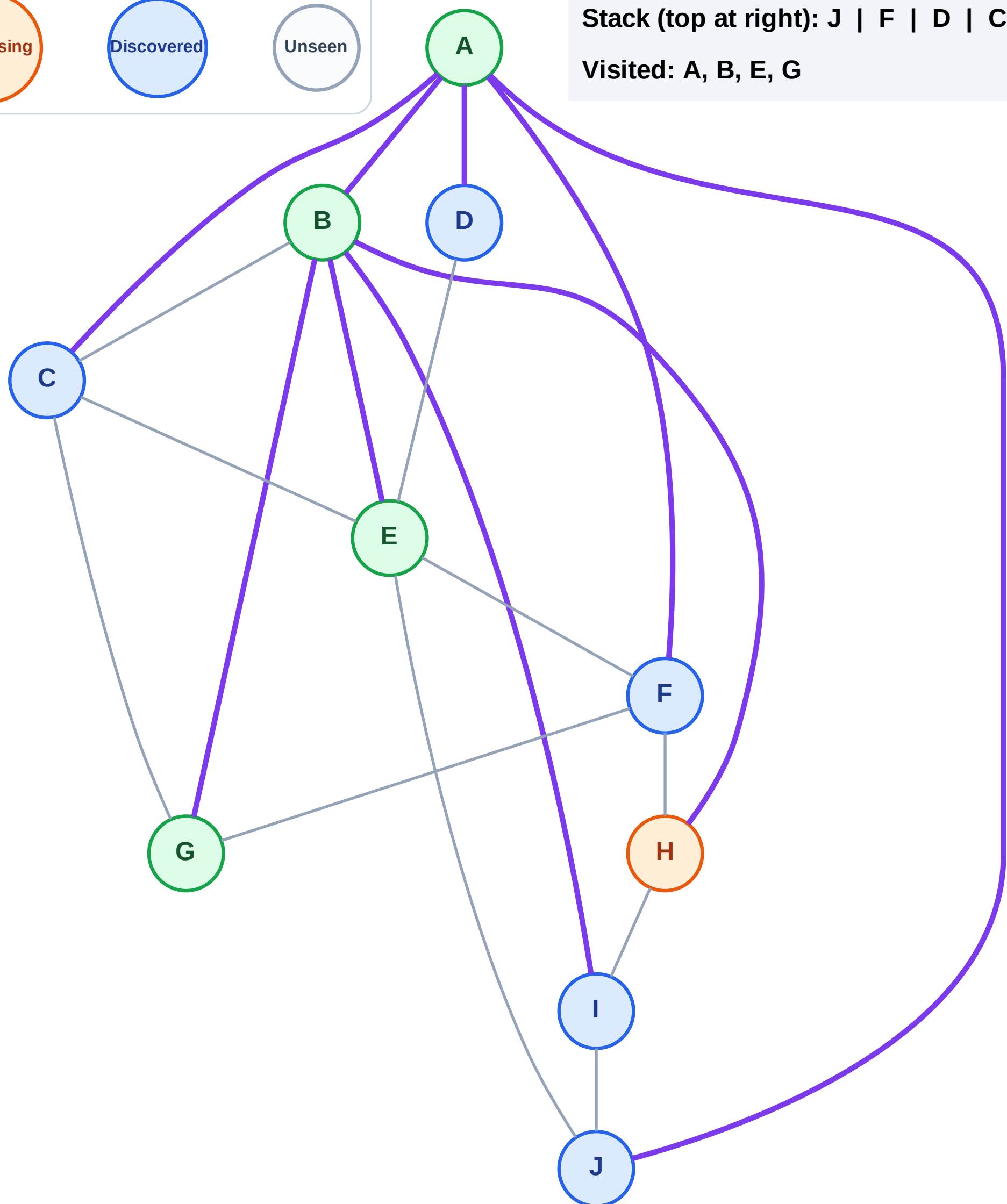
Step 10: Process node H

Legend



Stack (top at right): J | F | D | C | I

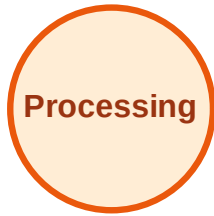
Visited: A, B, E, G



DFS Traversal

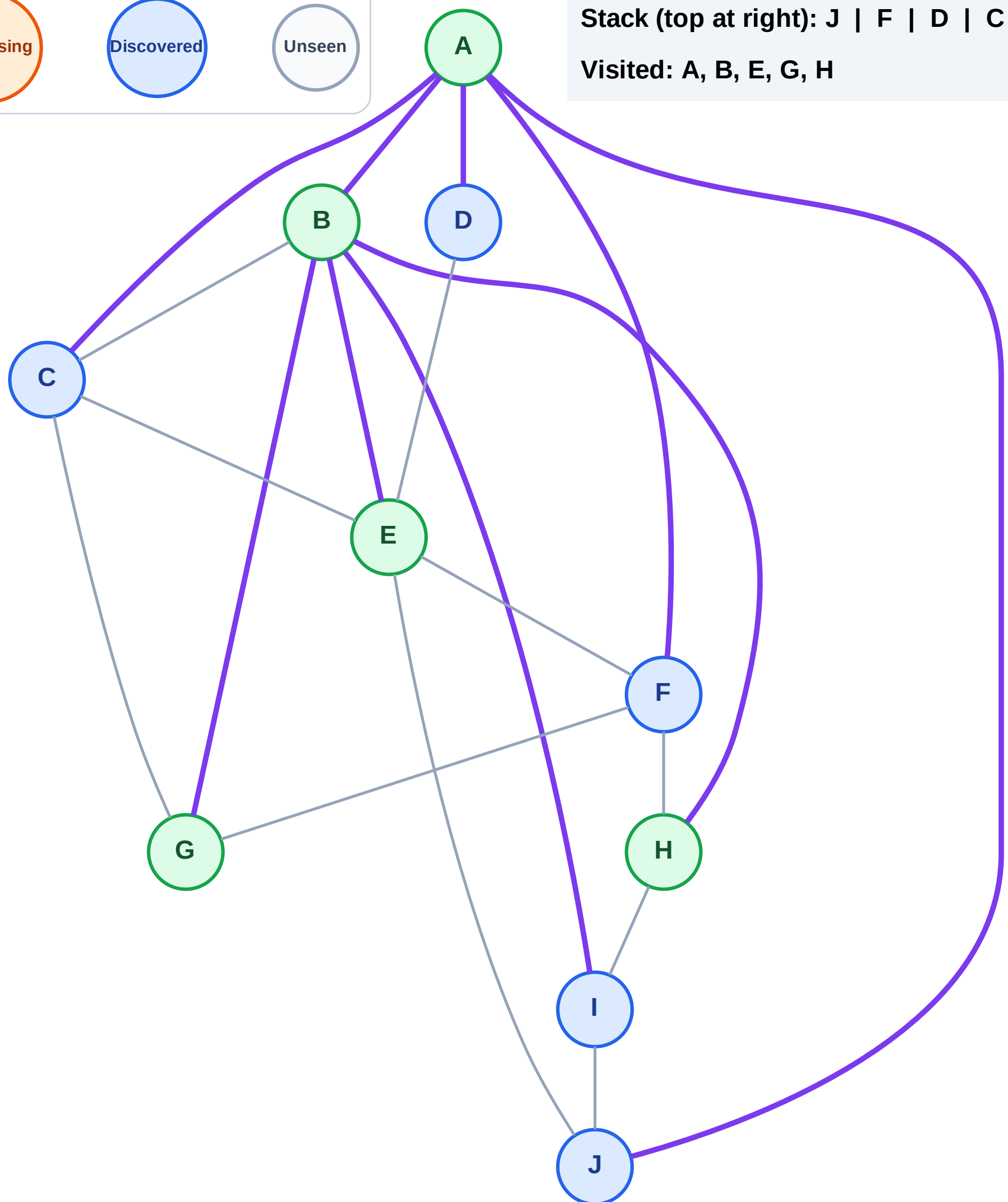
Step 11: No new neighbors from H

Legend



Stack (top at right): J | F | D | C | I

Visited: A, B, E, G, H



DFS Traversal

Step 12: Process node I

Legend

Complete

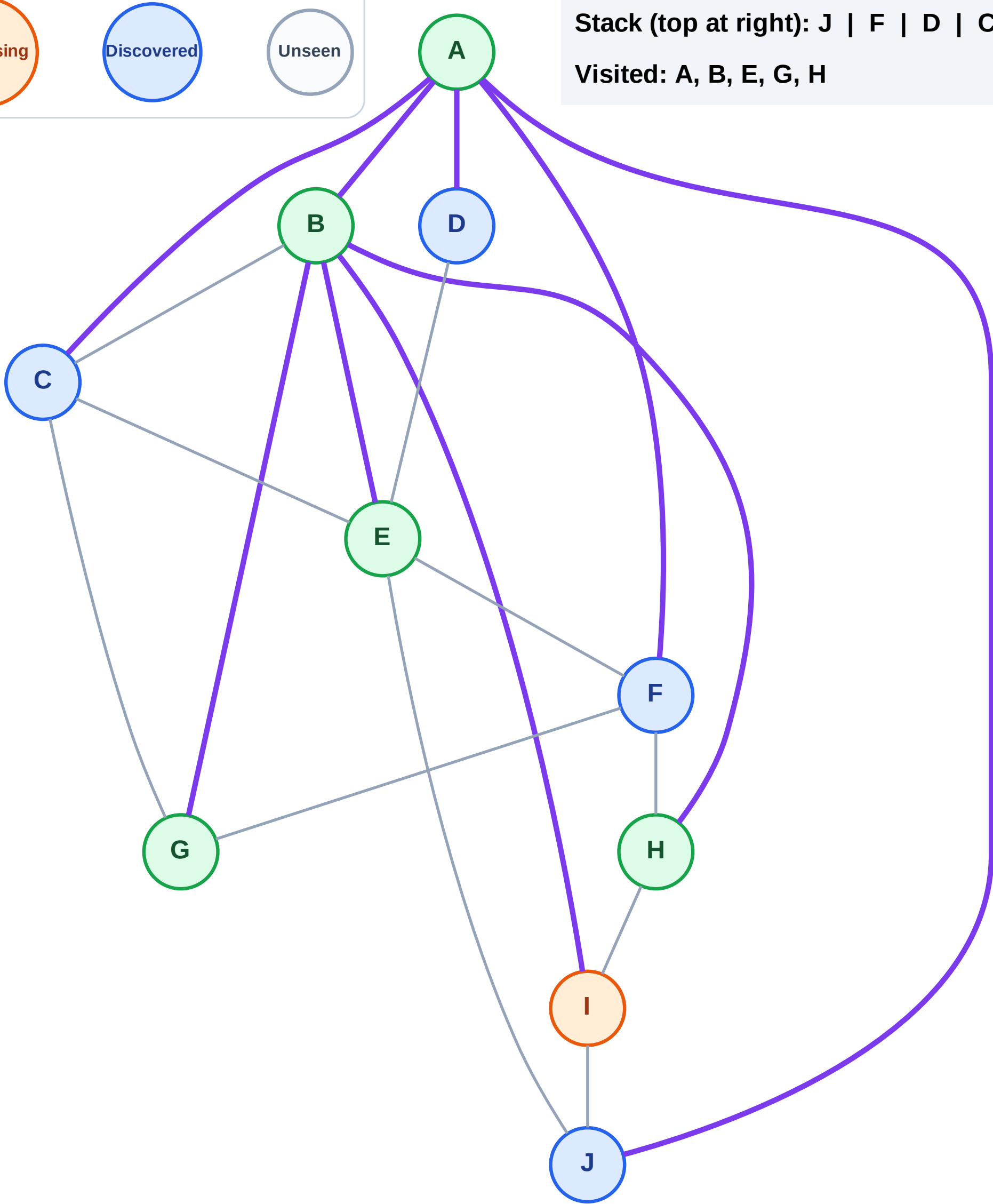
Processing

Discovered

Unseen

Stack (top at right): J | F | D | C

Visited: A, B, E, G, H



DFS Traversal

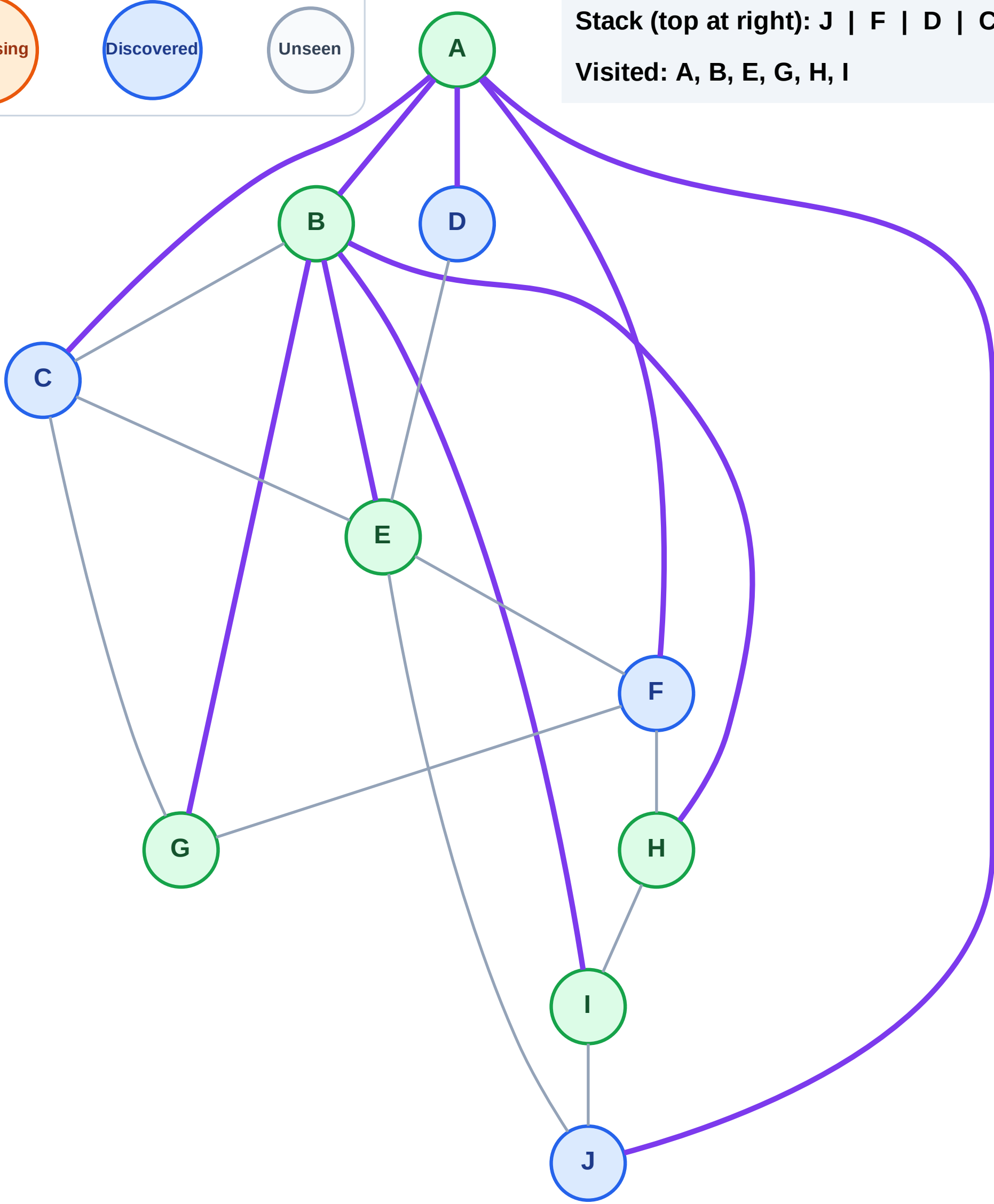
Step 13: No new neighbors from I

Legend



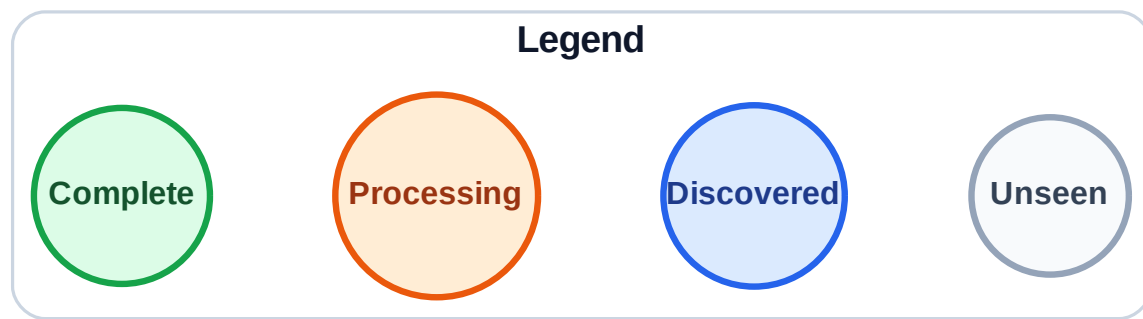
Stack (top at right): J | F | D | C

Visited: A, B, E, G, H, I

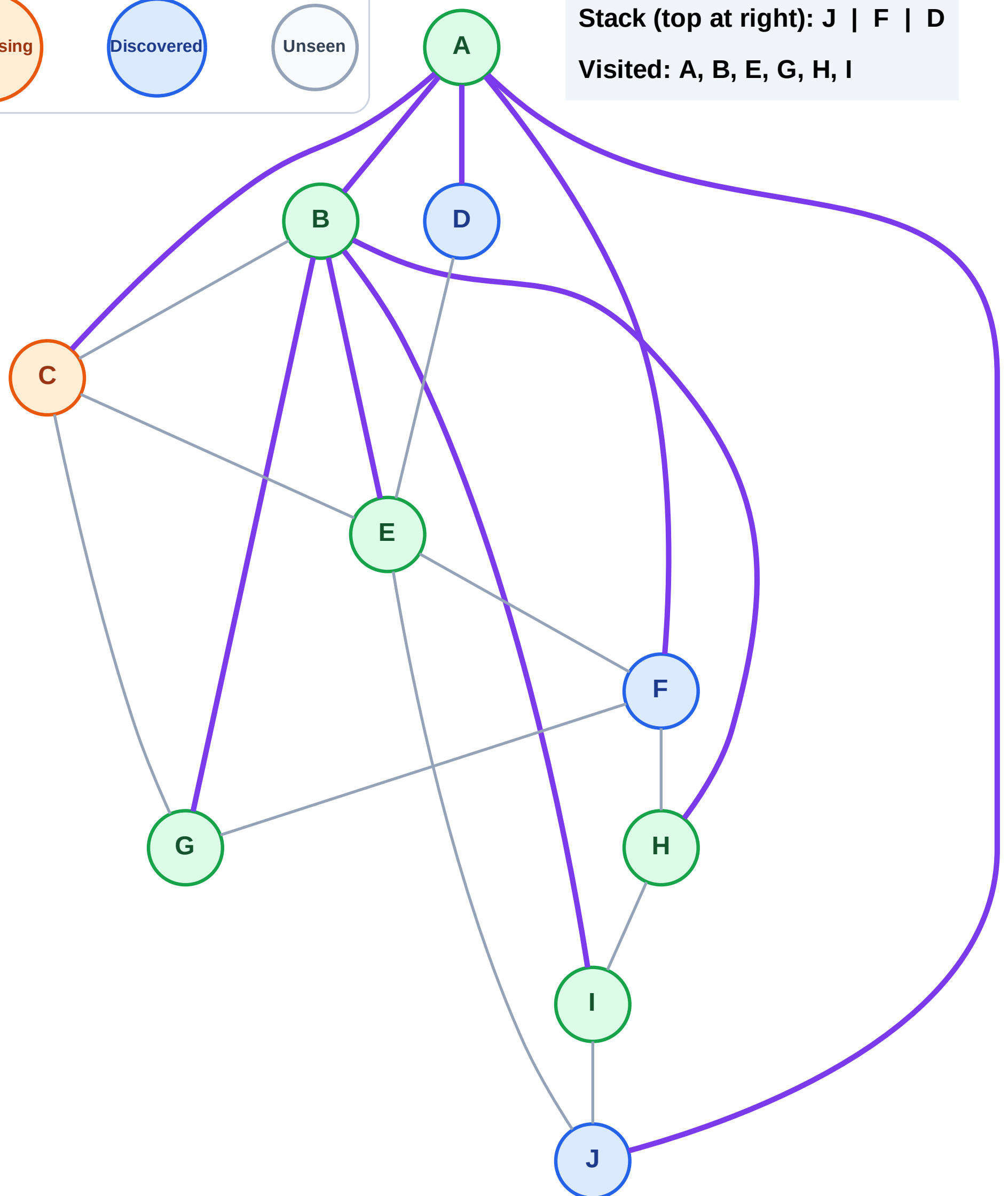


DFS Traversal

Step 14: Process node C



Stack (top at right): J | F | D
Visited: A, B, E, G, H, I



DFS Traversal

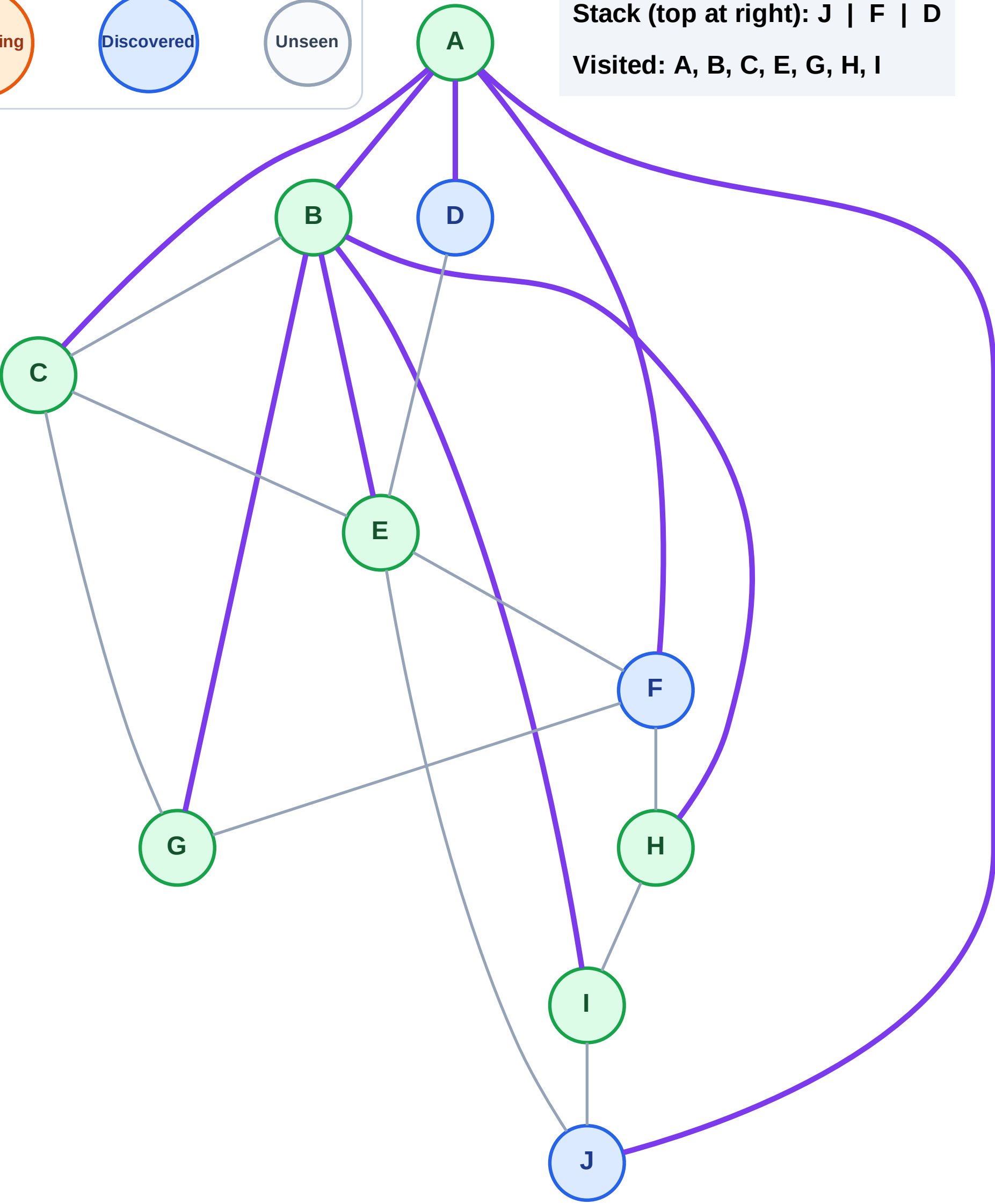
Step 15: No new neighbors from C

Legend



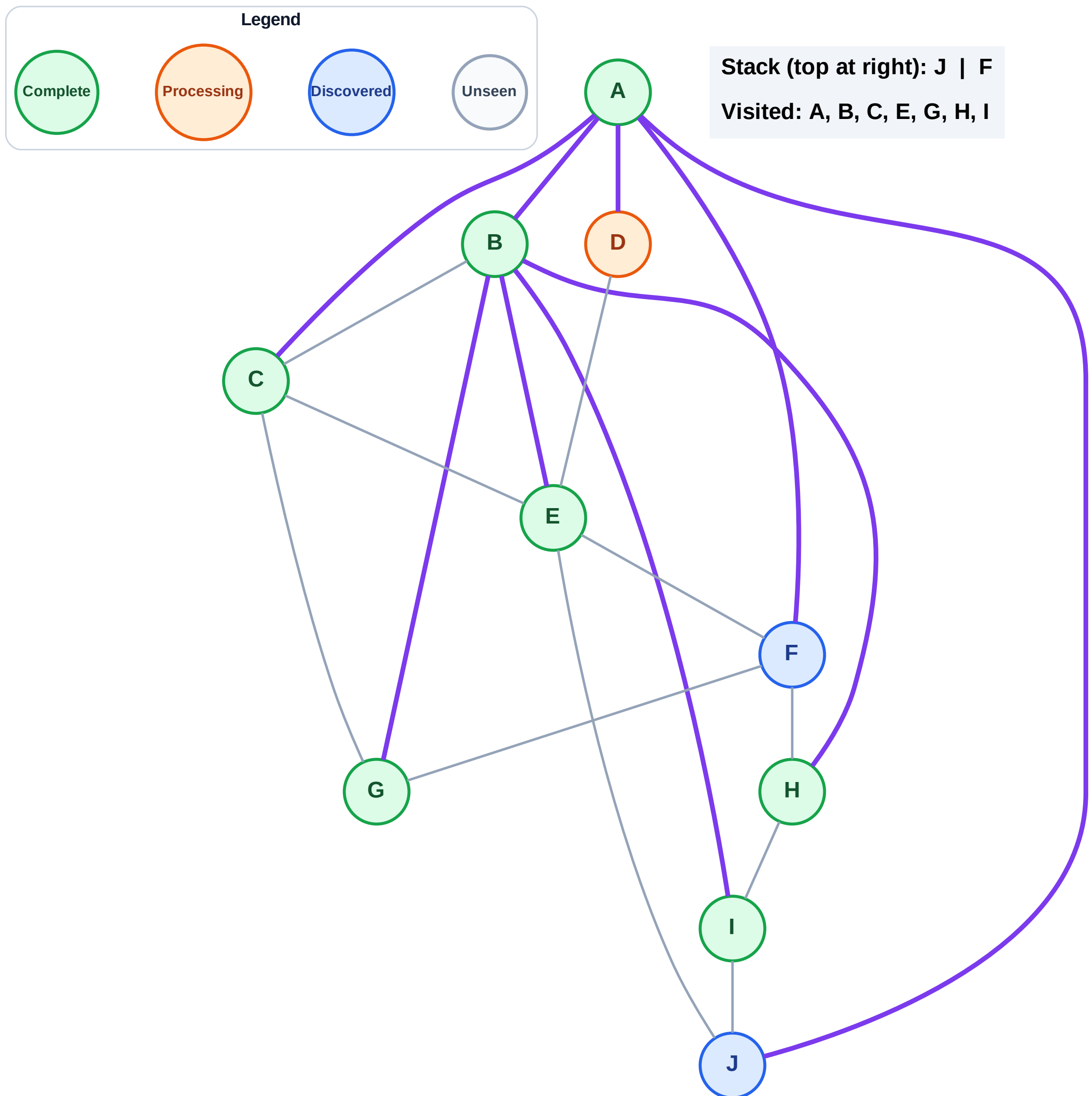
Stack (top at right): J | F | D

Visited: A, B, C, E, G, H, I



Step 16: Process node D

Step 16: Process node D



DFS Traversal

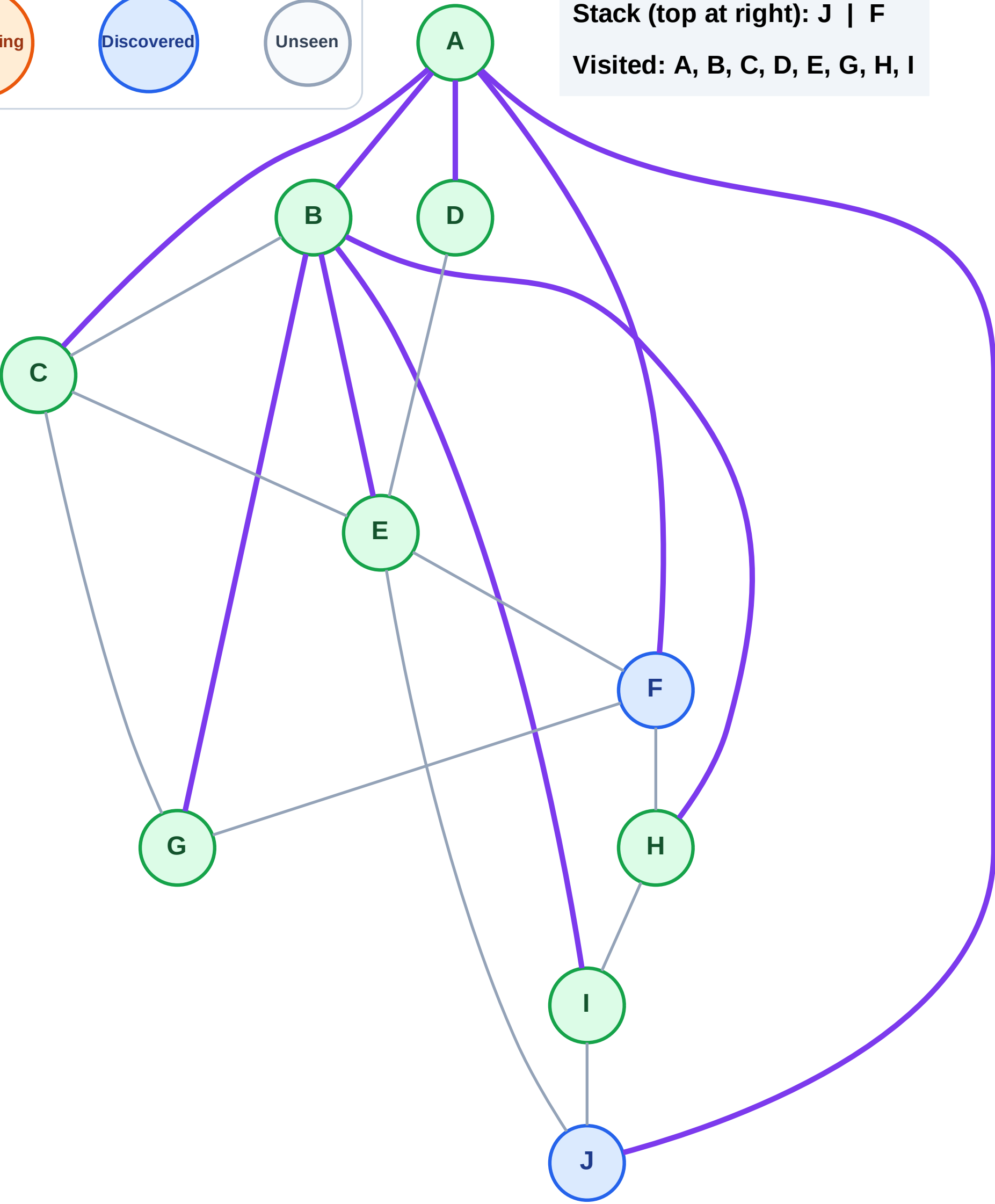
Step 17: No new neighbors from D

Legend



Stack (top at right): J | F

Visited: A, B, C, D, E, G, H, I



DFS Traversal

Step 18: Process node F

Legend

Complete

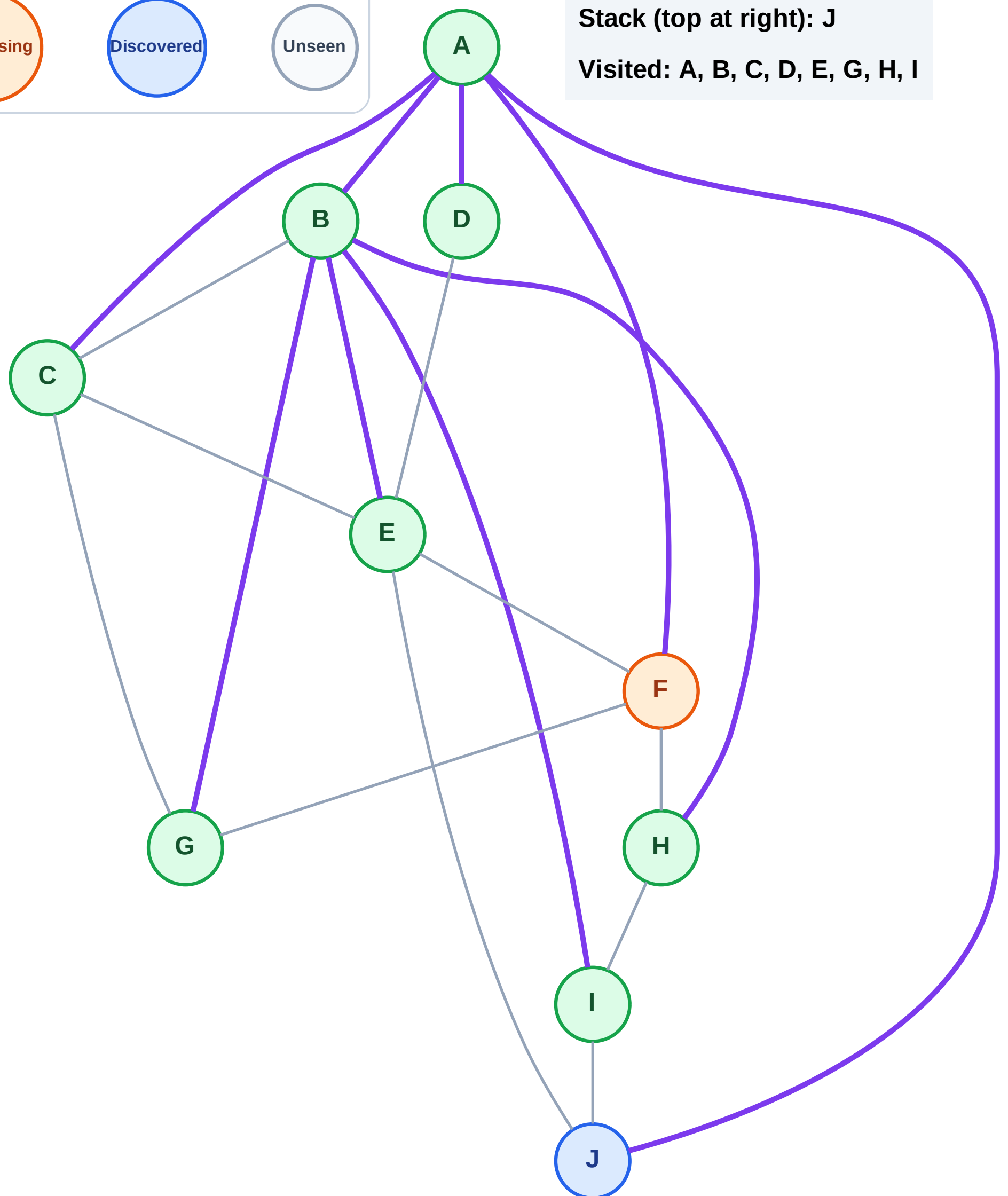
Processing

Discovered

Unseen

Stack (top at right): J

Visited: A, B, C, D, E, G, H, I



DFS Traversal

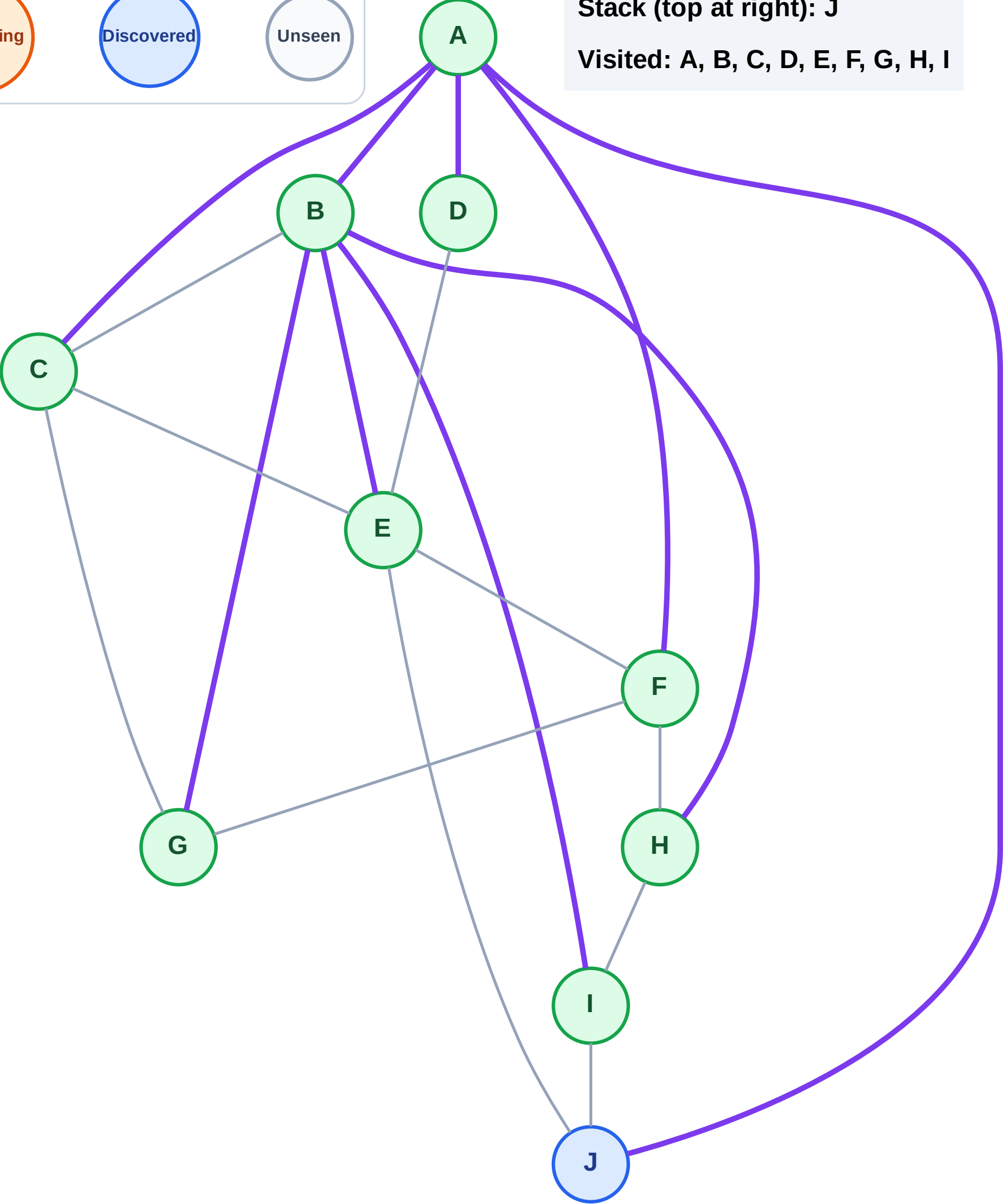
Step 19: No new neighbors from F

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

Complete

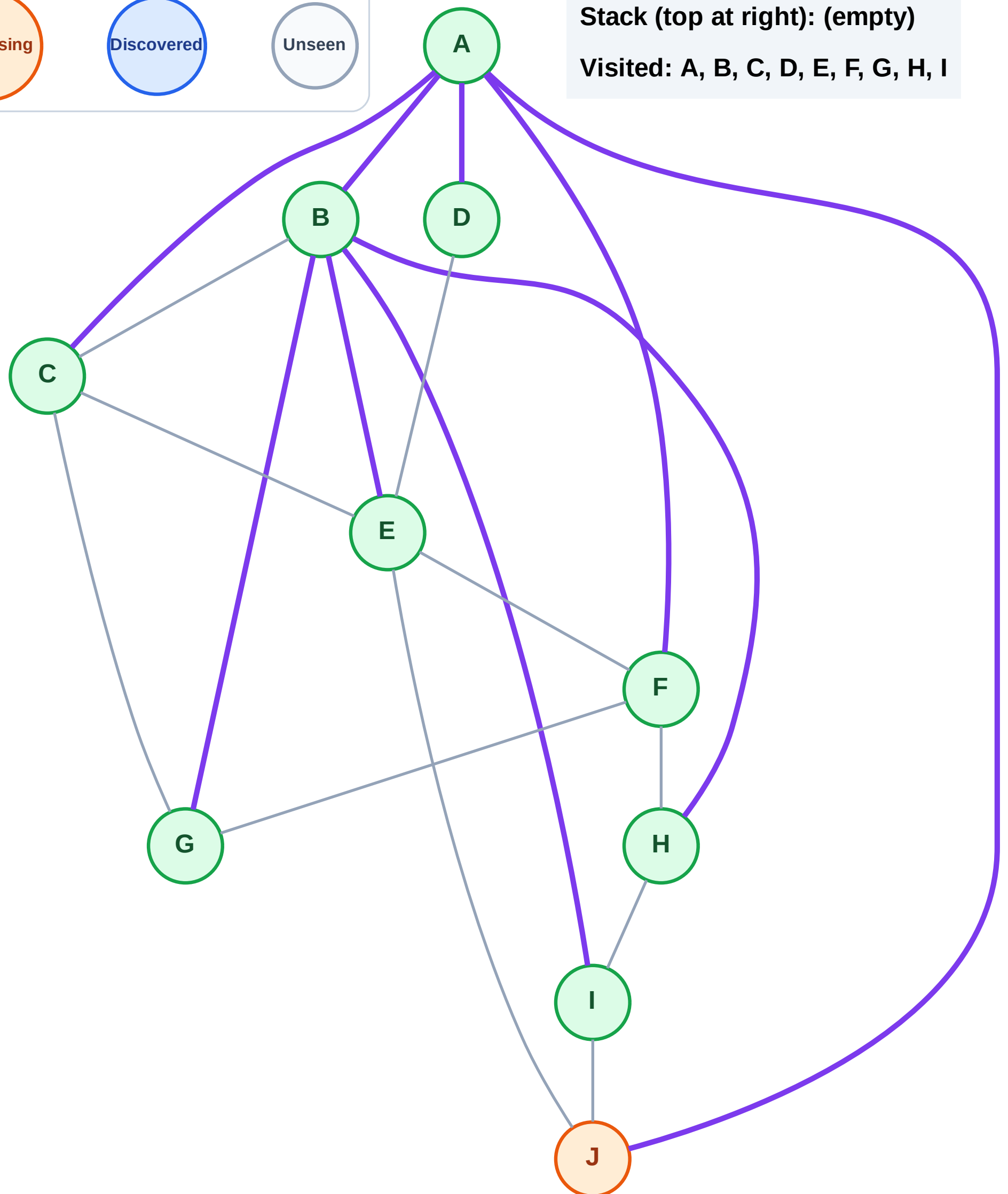
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

